WSQ • AUTOMATION MANAGEMENT IN PRODUCT DEVELOPMENT

Agentic AI for Product Development

Discover → design → deploy → learn

TGS-2024045799 | v7.0 | 2 days / 16 hours

Tertiary Infotech Academy Pte. Ltd.
UEN 201200696W

Mechanism-led courseware • Twelve self-contained activities

Digital attendance via TRAQOM

QR

Complete the attendance action before technical work begins.

1. **Scan the trainer-provided QR code**
2. Confirm the correct course code and session
3. Use your registered learner identity
4. Show the confirmation screen if requested

Trainer profile · General

?

Complete this profile before facilitating the agentic AI product-development course.

1. **Name:** ______________________
2. Title / designation: ______________________
3. Qualifications: ______________________
4. AI product and facilitation experience: ______________________

Dr Alfred Ang

AA

Principal Trainer, Tertiary Infotech Academy

1. **Doctorate-level applied technology educator**
2. AI, data, cloud and automation curriculum leadership
3. Industry-oriented assessment and implementation practice
4. Focus: explainable, governed technical outcomes

Learner introduction

YOU

Connect your existing role to one software or knowledge-work automation problem.

1. **Role and organisation context**
2. Current development or operations workflow
3. One repeated process bottleneck
4. One risk you want the agent to control

Working agreements

RULE

Use governed experimentation throughout the course.

1. **Use synthetic or approved data only**
2. Never paste live secrets into prompts or files
3. Keep consequential actions behind approval
4. Challenge completion claims with observable evidence

Course facts and competency context

TSC

ICT-TEM-4035-1.1 Automation Management in Product Development

1. **Course code: TGS-2024045799**
2. Duration: 2 days / 16 training hours
3. Assessment: 2 hours
4. Version: v7.0 / 11 September 2026

Day 1 lesson plan

D1

Evaluate opportunities and design evidence-grounded agentic product workflows.

1. **Product opportunity and JTBD framing**
2. Builder comparison and readiness
3. Research evidence and persona assumptions
4. Activities 01–06: discovery through AI-ready PRD

Day 2 lesson plan

D2

Optimise, deploy and govern an agentic product solution.

1. **Prompt contracts, evaluation sets and iteration**
2. Deployment architecture, security and human gates
3. KPI feedback loops and risk-adjusted decisions
4. Activities 07–12 plus 2-hour assessment

Learning outcomes

LO

At course completion, you can evaluate, optimise and govern an agentic product-development workflow.

1. **LO1 Evaluate agentic AI builders for product research and development, comparing strengths, limitations, resources and skills.**

2. LO2 Apply optimisation techniques to improve the efficiency, quality and control of agentic product-development workflows.

3. LO3 Evaluate the benefits and trade-offs of deploying agentic AI-powered product solutions and recommend a governed path forward.

Briefing for assessment

A

Assessment evidence must show your own decisions and a functioning workflow contract.

1. **Read every question and task before starting**
2. Use the supplied synthetic case and lab artifacts
3. Show requested repository, diff and test evidence
4. Do not cross an external or destructive boundary

Assessment and funding

$

Assessment is competence-based; course funding depends on current eligibility and conditions.

1 **Written Assessment: 60 minutes**

2 Practical Performance: 60 minutes

3 Course page lists support of up to 70% subject to eligibility

4 Check the current official course page before enrolment

Assessment flow

1→4

The assessor evaluates knowledge evidence, workflow performance and product decisions.

1. **Identity and instructions**
2. WA: K1–K6
3. PP: A1–A3
4. Outcome and follow-up

Download course material

↓

Use the LMS course record as the single verified entry point for learner materials.

1. **Sign in at lms-tms.tertiaryinfotech.com**
2. Open Agentic AI for Product Development
3. Select Course Material and confirm TGS-2024045799
4. Download slides, Learner Guide and activity files

https://lms-tms.tertiaryinfotech.com/

Practice exam portal

Q

Use only the practice or readiness activity assigned by the facilitator for this WSQ course.

1. **Open exams.tertiaryinfotech.com**
2. Sign in with the assigned learner account
3. Choose the facilitator-assigned readiness check
4. Review explanations and return to the assessed activity gaps

https://exams.tertiaryinfotech.com/

Access the activities

ACT

Twelve self-contained activity folders carry the complete end-to-end workflow.

1. **Open the individual activity README/PDF**
2. Use only the supplied mock data and YAML brief
3. Follow detailed steps in the Learner Guide
4. Record evidence in the folder checklist

https://github.com/tertiarycourses/TGS-2024045799-Agentic-AI-for-Product-Development

1

TOPIC 1

Evaluating Agentic AI Tools for Product Research and Development

Identify a product opportunity, map it to a suitable agentic pattern, and compare builders using evidence, readiness and total-cost criteria.

1 CONTROL

2 EVIDENCE

3 AUTOMATION

4 DECISION

Every product decision ends in observable evidence—not an agent’s assertion.

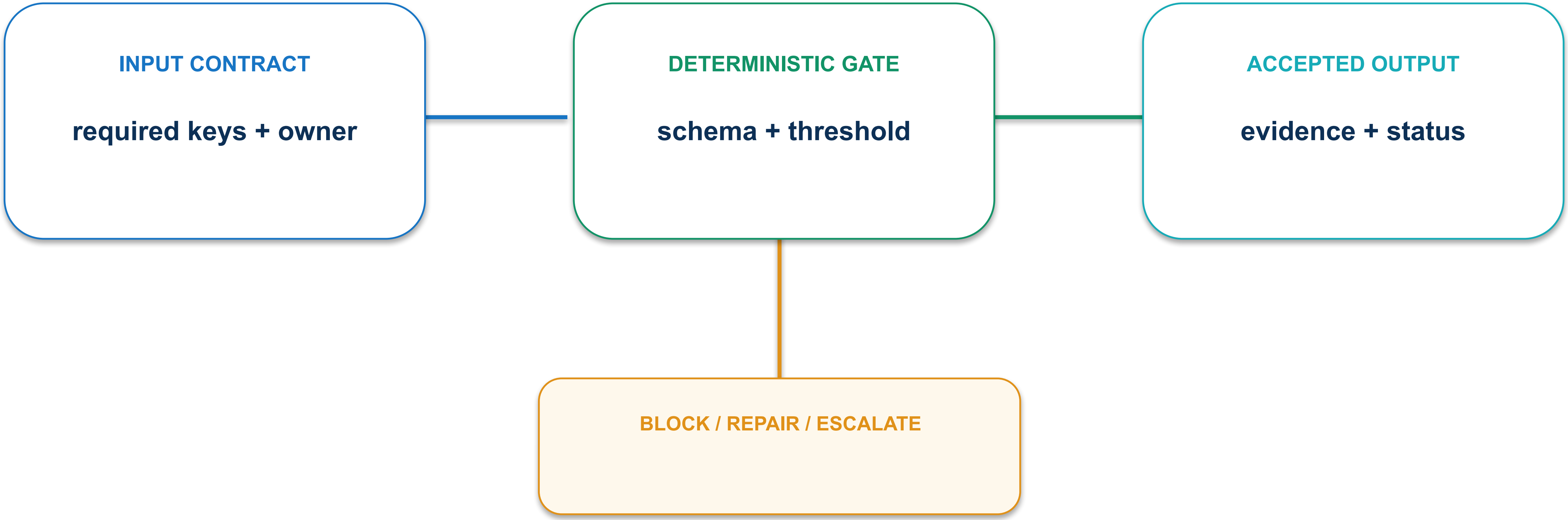
01 · Agentic AI versus assistants: execution path

AGENTIC PRODUCT LOOP

1	DISCOVER	Evidence and opportunity scope
2	DEFINE	Outcome, users and constraints
3	BUILD	Thin prototype and workflow
4	LEARN	Evaluation and next decision

CONCEPT VISUAL • LABELLED

01 · Agentic AI versus assistants: contract and control



Require a multi-step loop, tool boundary and stop condition before calling a system agentic.

01 · Agentic AI versus assistants: evidence and decision

MEASURE

Goal completion rate with valid evidence

ACCEPTANCE EVIDENCE

Trace showing plan, tool result and final decision

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

02 · Product lifecycle application map: execution path

1. **Discover need**
2. **Define concept**
3. **Deliver prototype**
4. **Learn in market**

STATE PASSED BETWEEN STAGES

insight

hypothesis

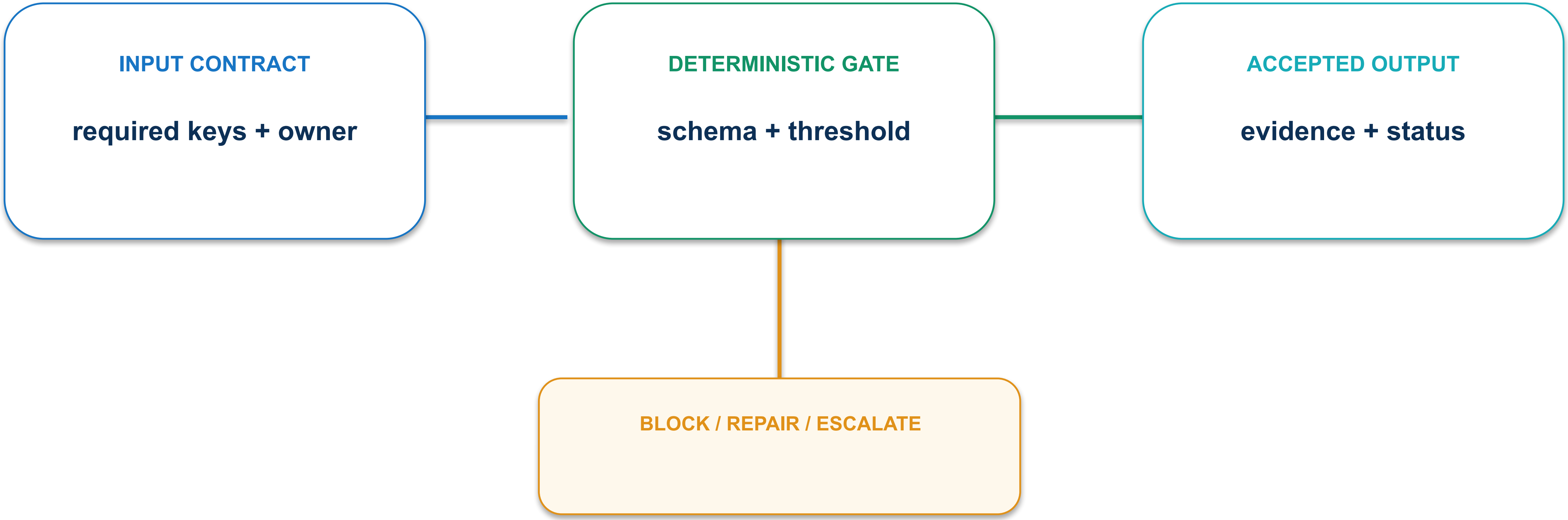
artifact

feedback

FAILURE MODE

Automation is applied everywhere without a clear decision or owner.

02 · Product lifecycle application map: contract and control



Map each agent task to one lifecycle decision and accountable role.

02 · Product lifecycle application map: evidence and decision

MEASURE

% lifecycle stages with a named human owner

ACCEPTANCE EVIDENCE

Lifecycle map with task-owner pairs

OBSERVED	THRESHOLD	DECISION	OWNER
machine value	approved rule	PASS / REPAIR	named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

03 · Opportunity framing: execution path

1 **Observe friction**

2 **Name user**

3 **Quantify impact**

4 **Frame outcome**

STATE PASSED BETWEEN STAGES

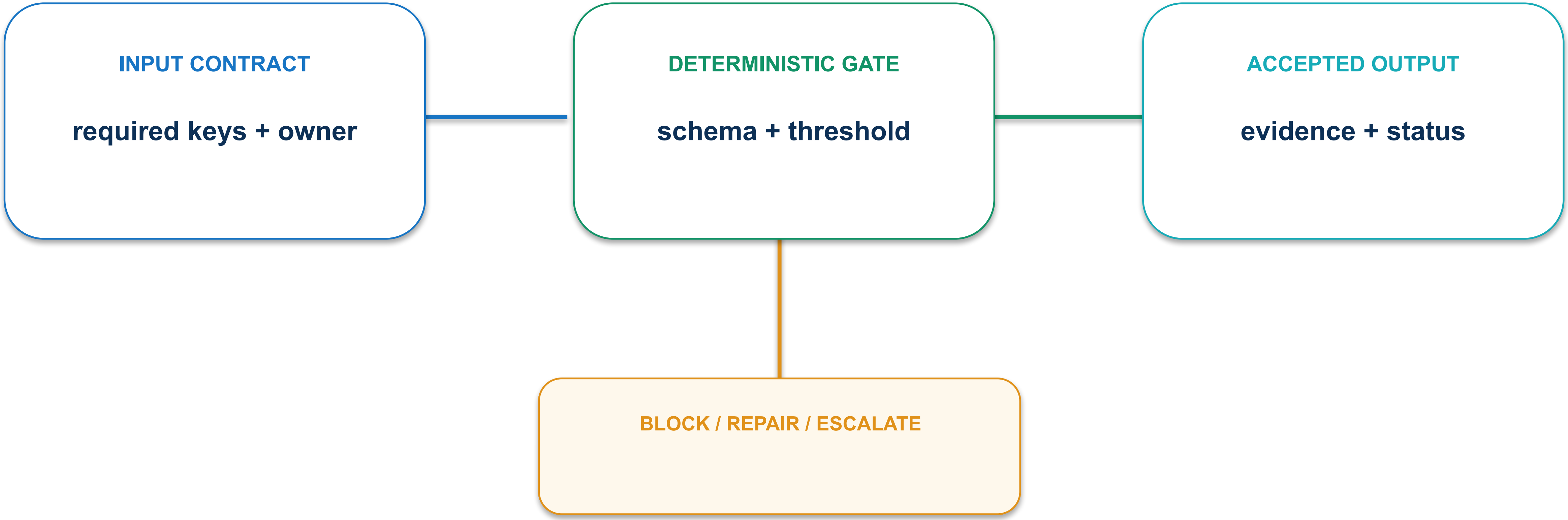
problem	**user**
baseline	**target**

FAILURE MODE

The team automates an interesting task that is not a material user problem.

CONTROL SURFACE

03 · Opportunity framing: contract and control



Start with observable friction and a measurable outcome, not a tool demonstration.

03 · Opportunity framing: evidence and decision

MEASURE

Validated problem statements / total ideas

ACCEPTANCE EVIDENCE

Opportunity brief with baseline and target

OBSERVED	THRESHOLD	DECISION	OWNER
machine value	approved rule	PASS / REPAIR	named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

MECHANISM

04 · Jobs-to-be-done lens: execution path

1. **Capture situation**
2. **Identify progress**
3. **Expose obstacle**
4. **Define success**

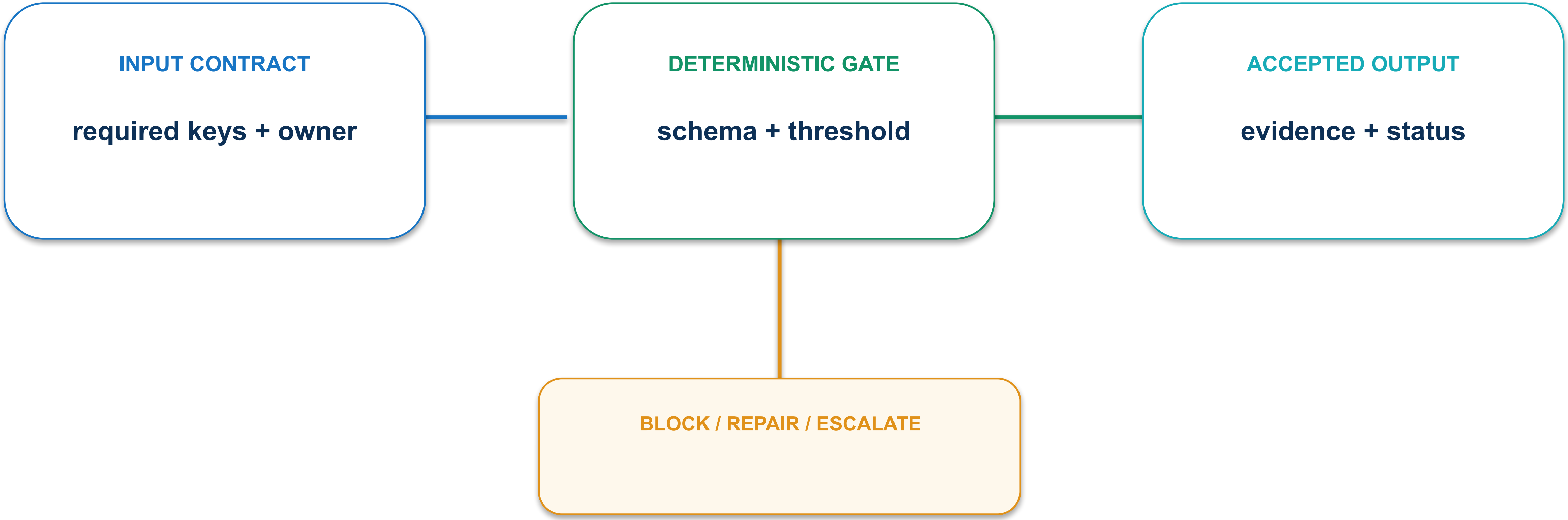
STATE PASSED BETWEEN STAGES

situation	**job**
barrier	**outcome**

FAILURE MODE

Personas become demographics without decision-relevant needs.

04 · Jobs-to-be-done lens: contract and control



Write needs as progress users seek in a real context.

04 · Jobs-to-be-done lens: evidence and decision

MEASURE

% research observations linked to a user job

ACCEPTANCE EVIDENCE

JTBD statement linked to verbatim research notes

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

MECHANISM

05 · Research-agent pattern: execution path

1. **Set question**
2. **Gather sources**
3. **Triangulate claims**
4. **Synthesize insight**

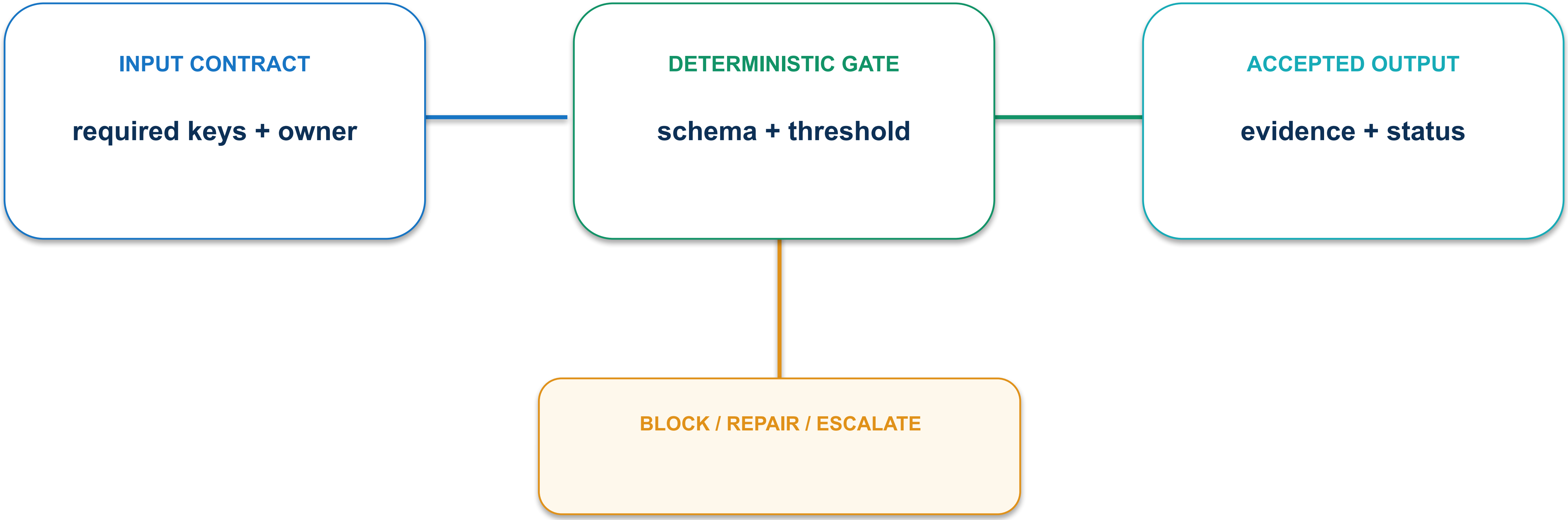
STATE PASSED BETWEEN STAGES

question	**source**
claim	**confidence**

FAILURE MODE

The agent produces polished synthesis from weak or circular evidence.

05 · Research-agent pattern: contract and control



Require source provenance, claim-level support and uncertainty labels.

05 · Research-agent pattern: evidence and decision

MEASURE

% material claims supported by two independent sources

ACCEPTANCE EVIDENCE

Research ledger and cited insight brief

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

06 · Competitor intelligence workflow: execution path

1 Choose peers

2 Freeze criteria

3 Collect evidence

4 Compare gaps

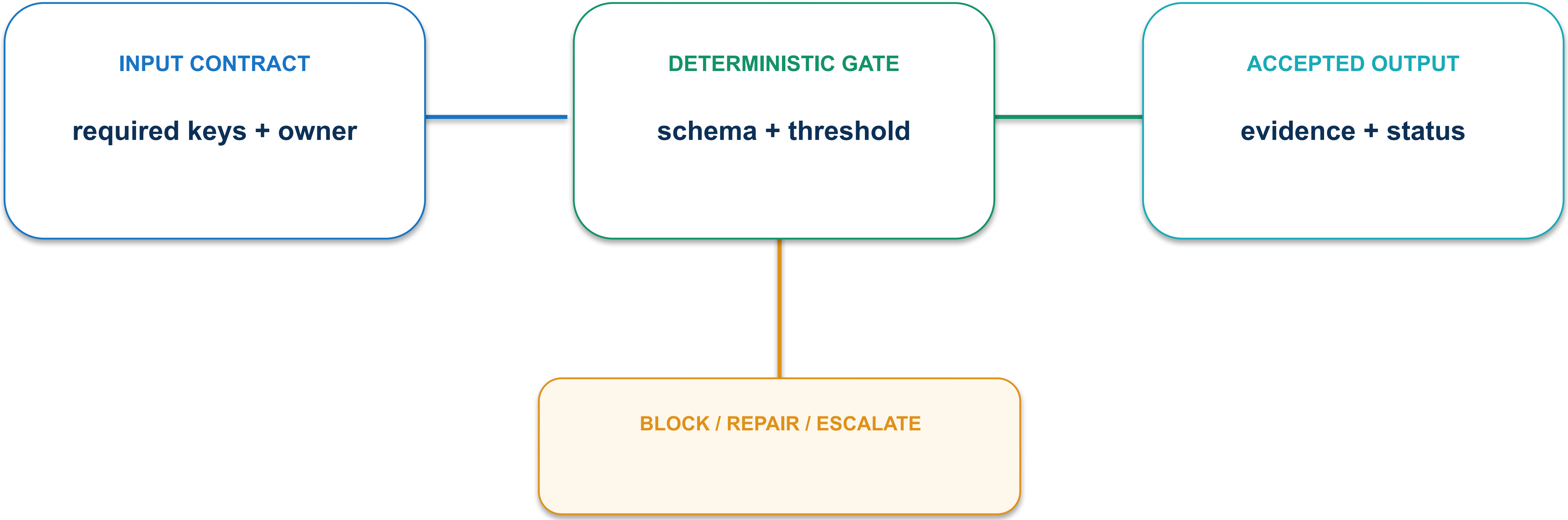
STATE PASSED BETWEEN STAGES

peer	**criterion**
evidence	**gap**

FAILURE MODE

Different evidence standards make the comparison misleading.

06 · Competitor intelligence workflow: contract and control



Freeze the same criteria and observation date before collection.

06 · Competitor intelligence workflow: evidence and decision

MEASURE

Comparable criteria populated per competitor

ACCEPTANCE EVIDENCE

Timestamped comparison matrix

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

07 · Voice-of-customer synthesis: execution path

1 **Ingest feedback**

2 **Cluster themes**

3 **Test outliers**

4 **Prioritise need**

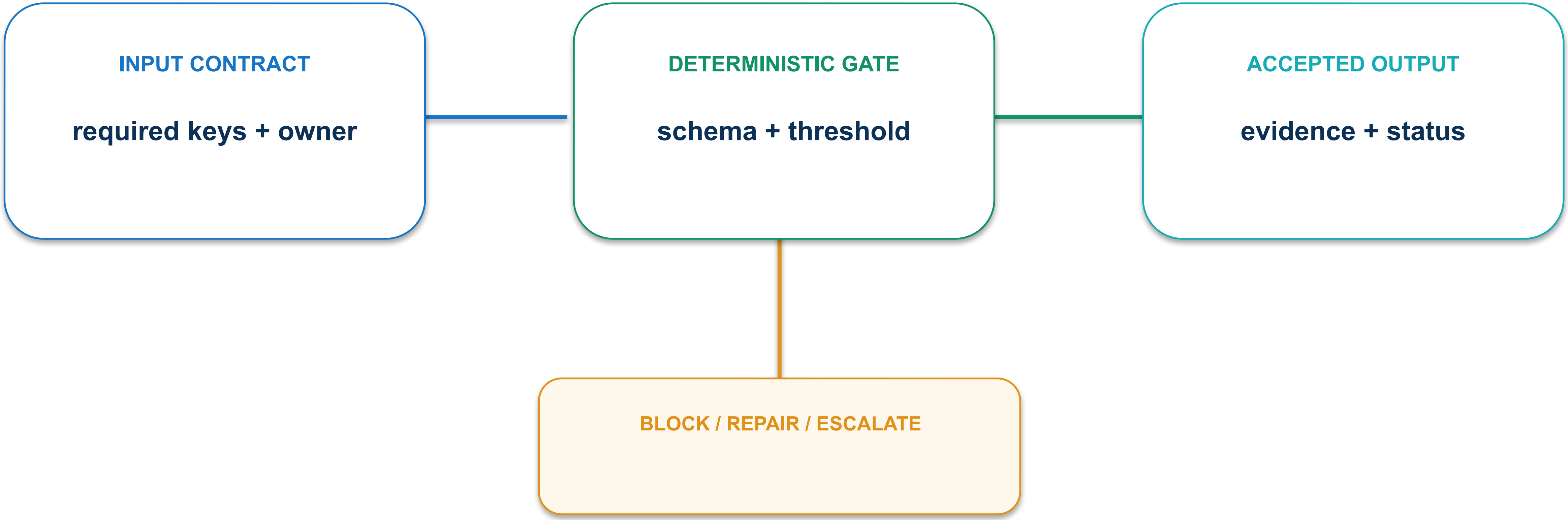
STATE PASSED BETWEEN STAGES

feedback_id	**theme**
outlier	**priority**

FAILURE MODE

A dominant-sounding summary hides minority but critical needs.

07 · Voice-of-customer synthesis: contract and control



Retain row-level traceability and inspect negative cases separately.

07 · Voice-of-customer synthesis: evidence and decision

MEASURE

% themes traceable to source rows

ACCEPTANCE EVIDENCE

Theme table with supporting and contradicting excerpts

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

08 · Persona and assumption map: execution path

1 **Group evidence**

2 **Draft persona**

3 **Label assumptions**

4 **Plan validation**

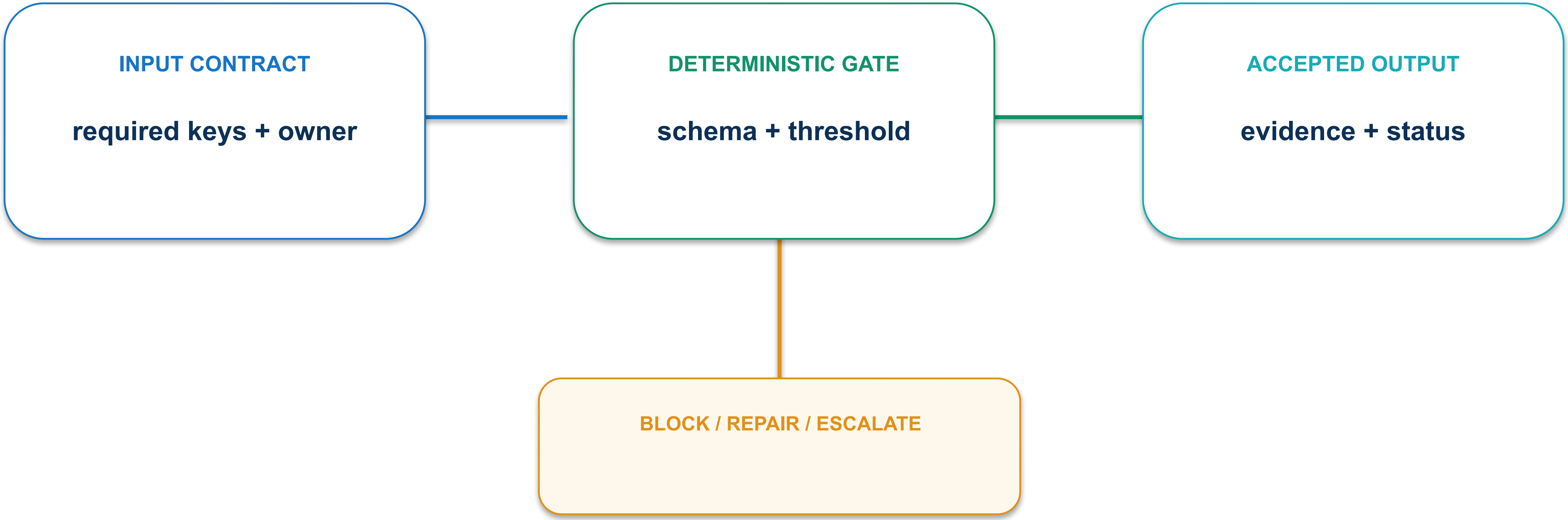
STATE PASSED BETWEEN STAGES

segment	**need**
assumption	**test**

FAILURE MODE

Generated personas are treated as facts about real users.

08 · Persona and assumption map: contract and control



Mark inferred attributes and separate evidence from hypothesis.

08 · Persona and assumption map: evidence and decision

MEASURE

High-risk assumptions with a validation test

ACCEPTANCE EVIDENCE

Persona card plus assumption register

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

MECHANISM

09 · Agentic builder landscape: execution path

1. Define use case
2. Shortlist builders
3. Run common task
4. Score fit

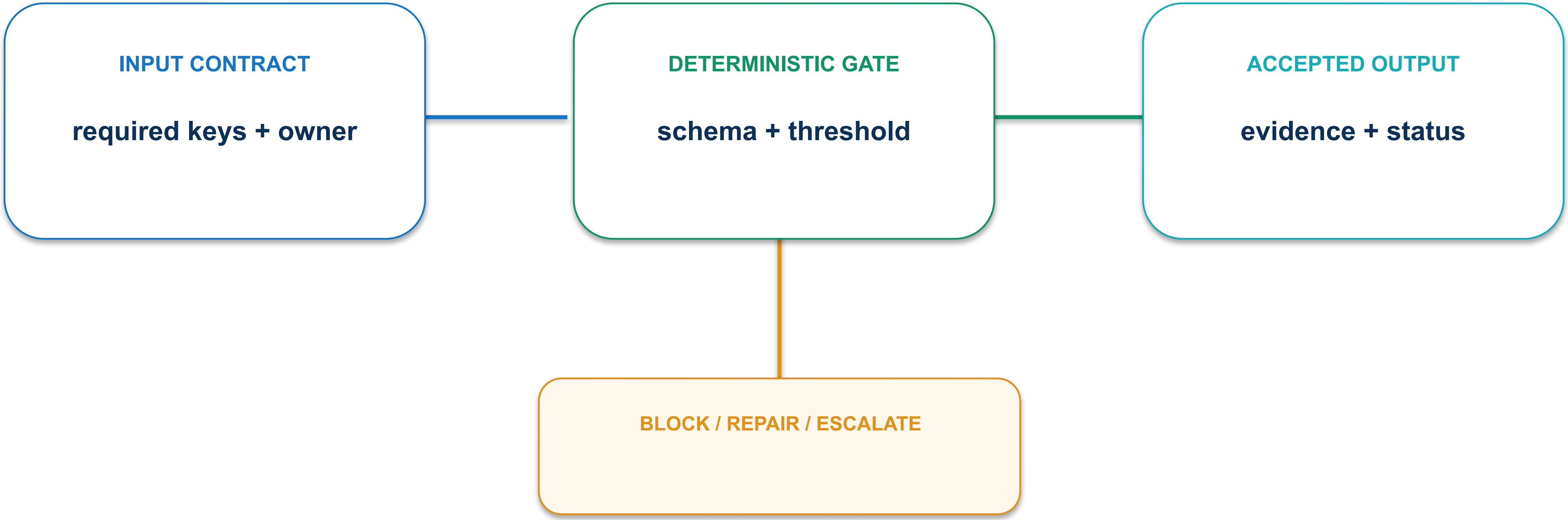
STATE PASSED BETWEEN STAGES

use_case	builder
test_case	score

FAILURE MODE

Feature lists replace direct comparison on the intended task.

09 · Agentic builder landscape: contract and control



Run the same input, output and constraints across every candidate.

09 · Agentic builder landscape: evidence and decision

MEASURE

Weighted score on a common golden task

ACCEPTANCE EVIDENCE

Paired trial results and recommendation

OBSERVED	THRESHOLD	DECISION	OWNER
machine value	approved rule	PASS / REPAIR	named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

MECHANISM

10 · Workflow versus autonomous agent: execution path

1. **Assess variability**
2. **Assess risk**
3. **Choose pattern**
4. **Set override**

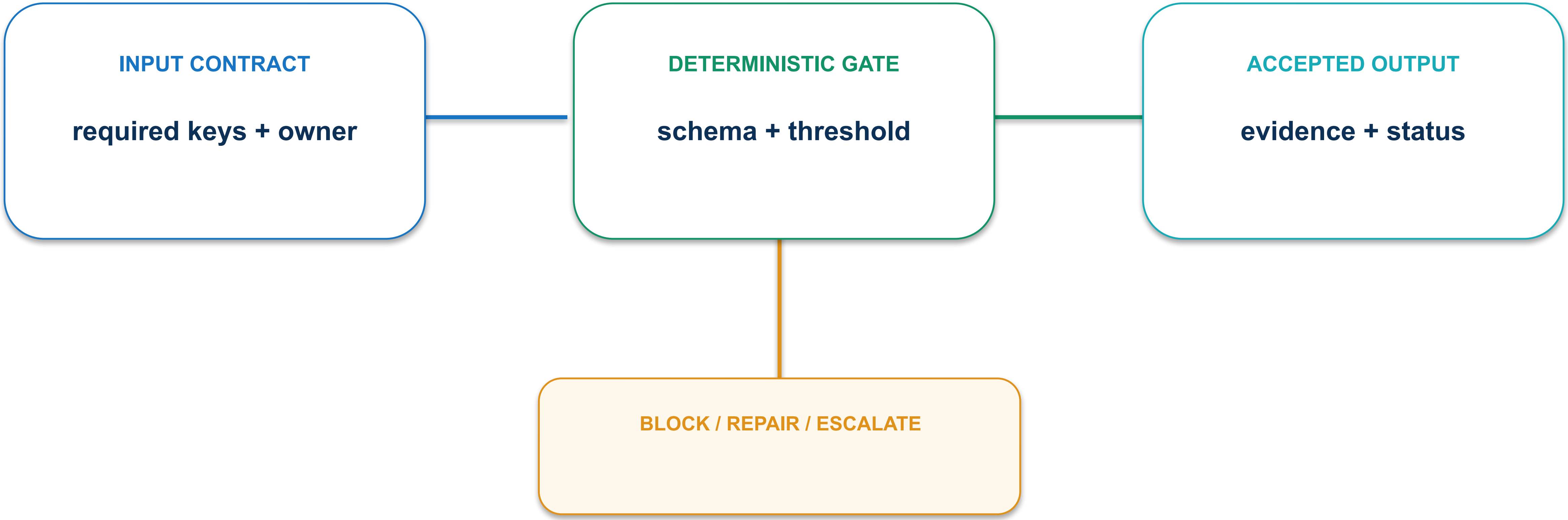
STATE PASSED BETWEEN STAGES

variability	**risk**
pattern	**override**

FAILURE MODE

Autonomy is added where a predictable workflow would be safer and cheaper.

10 · Workflow versus autonomous agent: contract and control



Use the simplest architecture that satisfies variability and judgment needs.

10 · Workflow versus autonomous agent: evidence and decision

MEASURE

% high-risk steps on deterministic paths

ACCEPTANCE EVIDENCE

Pattern decision with rejected alternatives

OBSERVED	THRESHOLD	DECISION	OWNER
machine value	approved rule	PASS / REPAIR	named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

MECHANISM

11 · Build-buy-partner decision: execution path

1 Clarify advantage

2 Estimate capability

3 Model ownership

4 Select posture

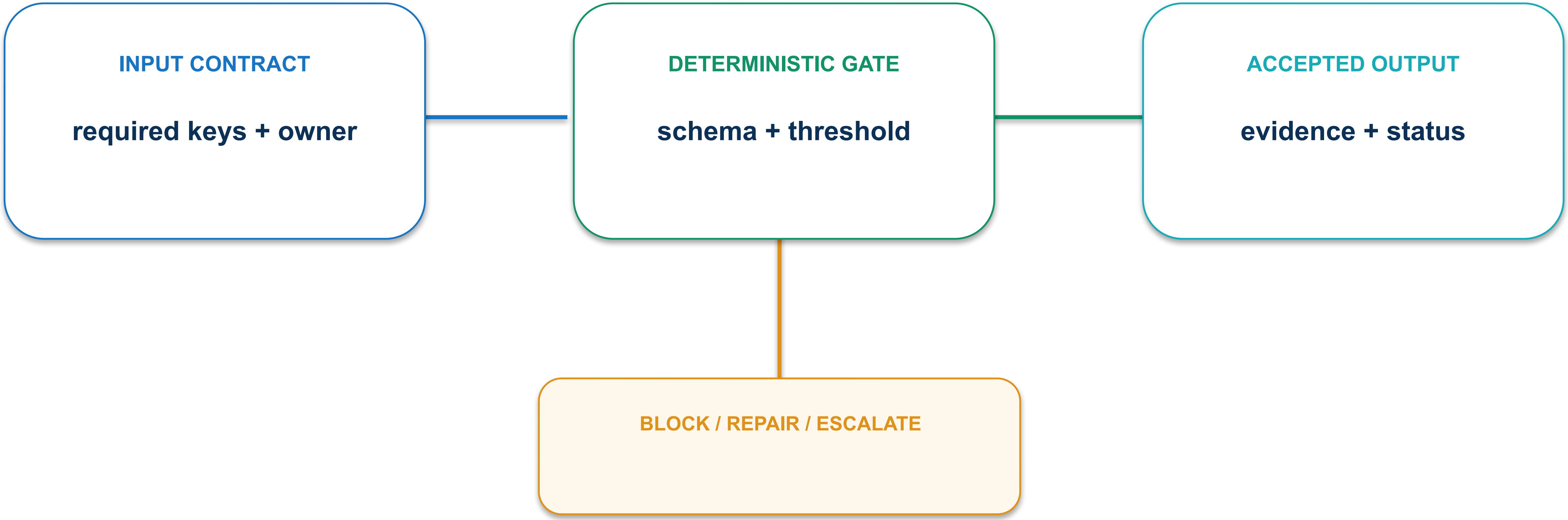
STATE PASSED BETWEEN STAGES

advantage	capability
ownership_cost	posture

FAILURE MODE

Subscription price is compared with in-house build cost while integration and governance are ignored.

11 · Build-buy-partner decision: contract and control



Include change, integration, assurance and exit costs.

11 · Build-buy-partner decision: evidence and decision

MEASURE

Risk-adjusted value over total cost of ownership

ACCEPTANCE EVIDENCE

Three-option decision table with sensitivity range

OBSERVED	THRESHOLD	DECISION	OWNER
machine value	approved rule	PASS / REPAIR	named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

MECHANISM

12 · Resource and skills readiness: execution path

1. List capabilities
2. Assess gaps
3. Plan controls
4. Sequence pilot

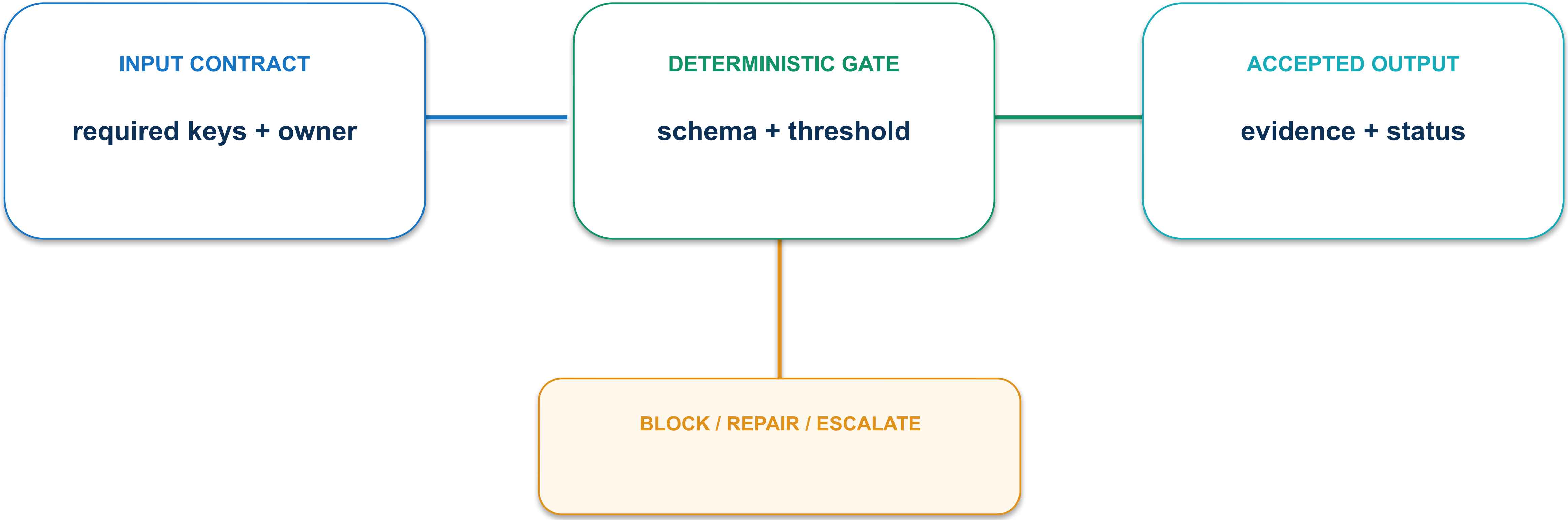
STATE PASSED BETWEEN STAGES

data	people
platform	governance

FAILURE MODE

A tool is selected before data access, ownership and evaluation skills exist.

12 · Resource and skills readiness: contract and control



Gate pilot entry on minimum data, product, technical and risk capabilities.

12 · Resource and skills readiness: evidence and decision

MEASURE

Critical readiness gaps closed before pilot

ACCEPTANCE EVIDENCE

Readiness heatmap and action owners

OBSERVED	THRESHOLD	DECISION	OWNER
machine value	approved rule	PASS / REPAIR	named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

MECHANISM

13 · Evaluation decision matrix: execution path

1 **Weight criteria**

2 **Score evidence**

3 **Test sensitivity**

4 **Recommend**

STATE PASSED BETWEEN STAGES

weight

score

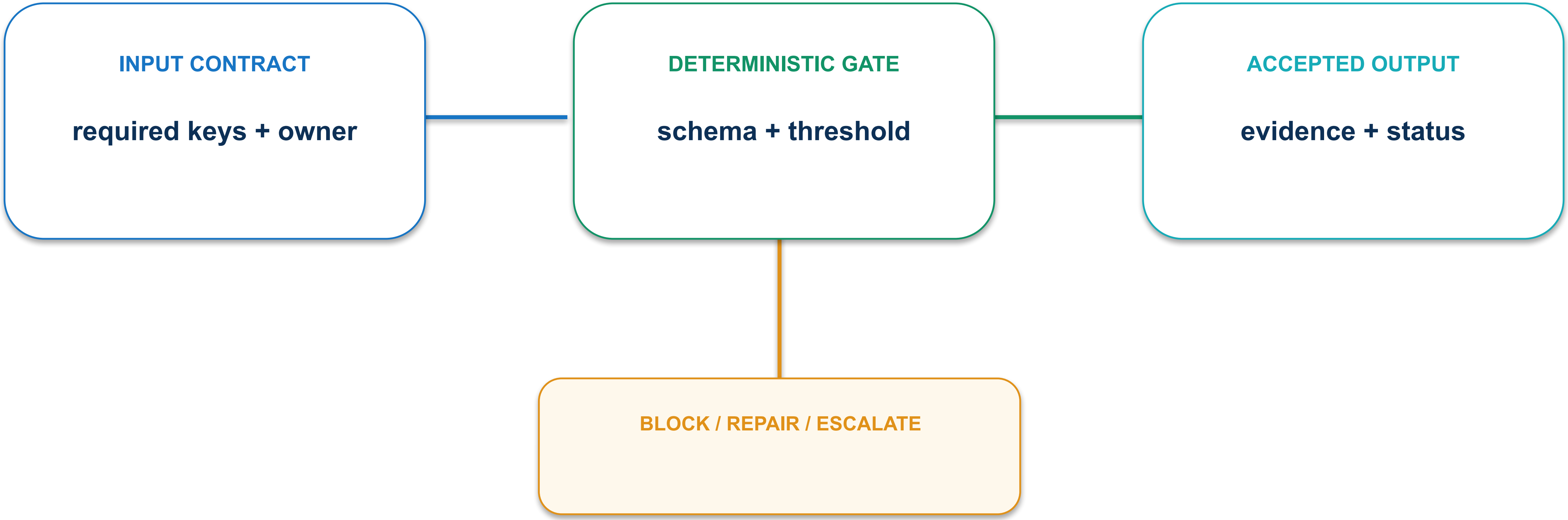
sensitivity

decision

FAILURE MODE

A predetermined winner is justified with arbitrary weights.

13 · Evaluation decision matrix: contract and control



Agree criteria before testing and show how weight changes affect the result.

13 · Evaluation decision matrix: evidence and decision

MEASURE

Recommendation stability across plausible weights

ACCEPTANCE EVIDENCE

Weighted matrix, caveats and decision owner

OBSERVED	THRESHOLD	DECISION	OWNER
machine value	approved rule	PASS / REPAIR	named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

Discovery boundary: evidence, agent synthesis and accountable human

1

RESEARCH

Collect and triangulate evidence

2

AGENT

Synthesize and expose uncertainty

3

PRODUCT

Prioritise need and own decision

Opportunity state: OBSERVED, FRAMED, TESTED and PRIORITISED

1 — OBSERVED

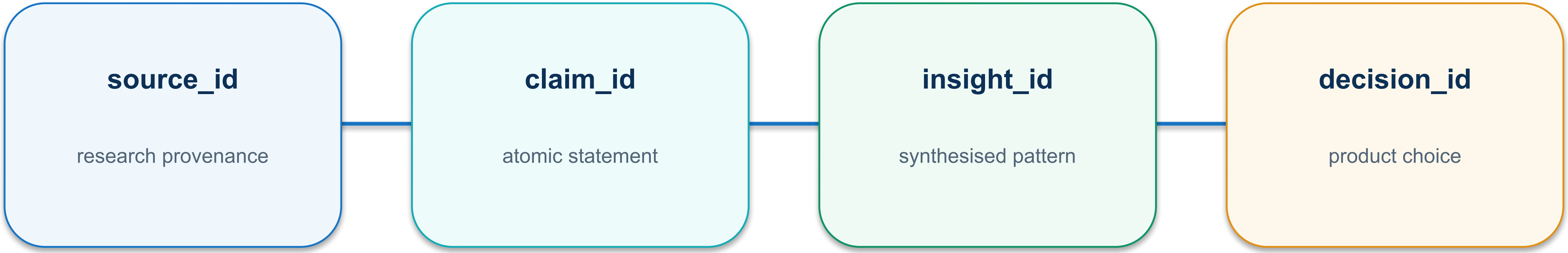
2 — FRAMED

3 — TESTED

4 — PRIORITISED

PRIORITISED means the opportunity is supported by evidence, not presentation polish.

Research evidence chain: source → claim → insight → decision



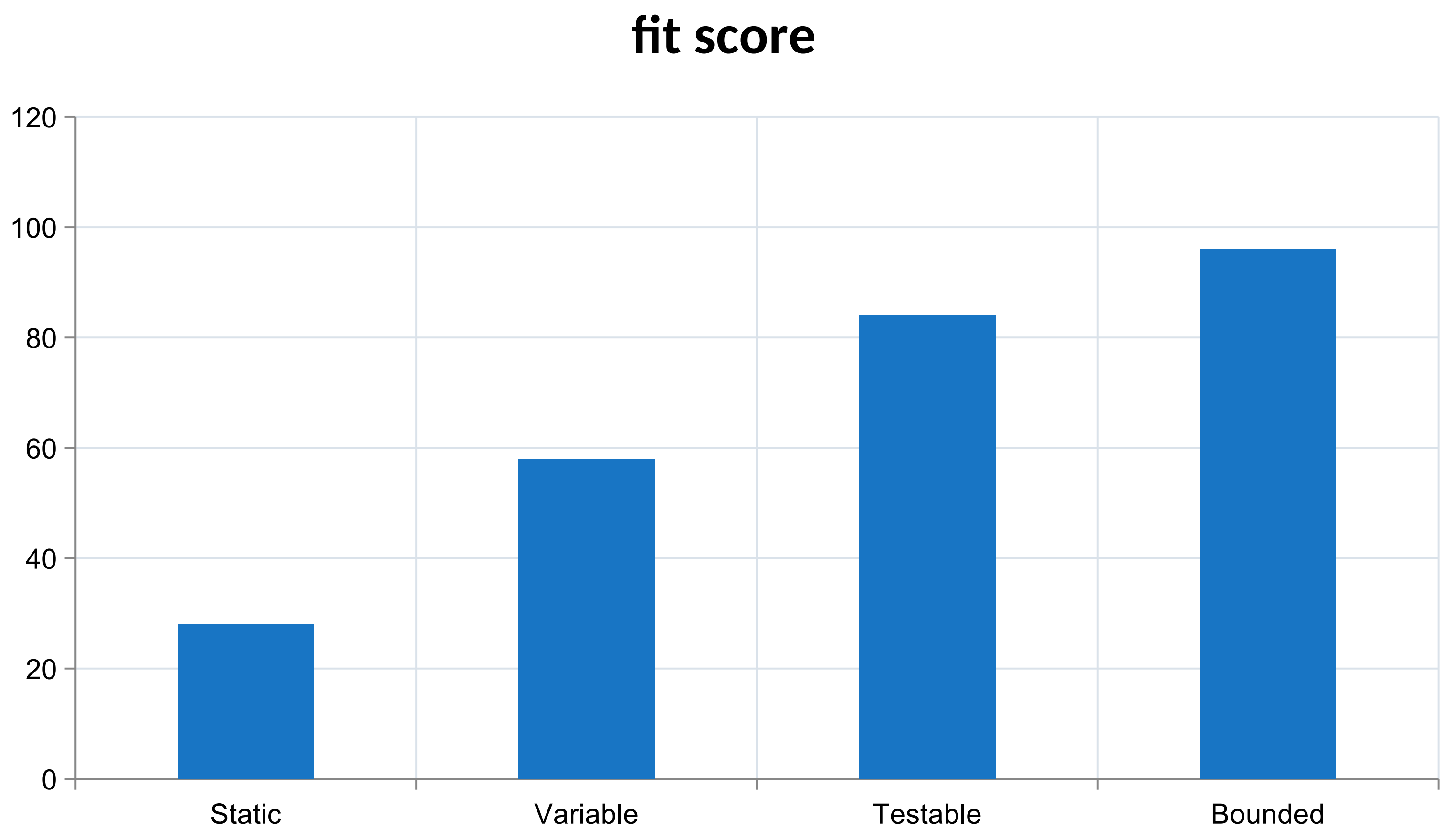
A discovery decision is explainable when each insight resolves to source evidence.

Discovery failure taxonomy: bias, gaps, conflict and overclaim

BIAS	broaden sampling and segment results
GAP	collect missing evidence
CONFLICT	retain competing explanations
OVERCLAIM	lower confidence and validate

EDITABLE NATIVE CHART

Agent fit rises with variability and verifiability



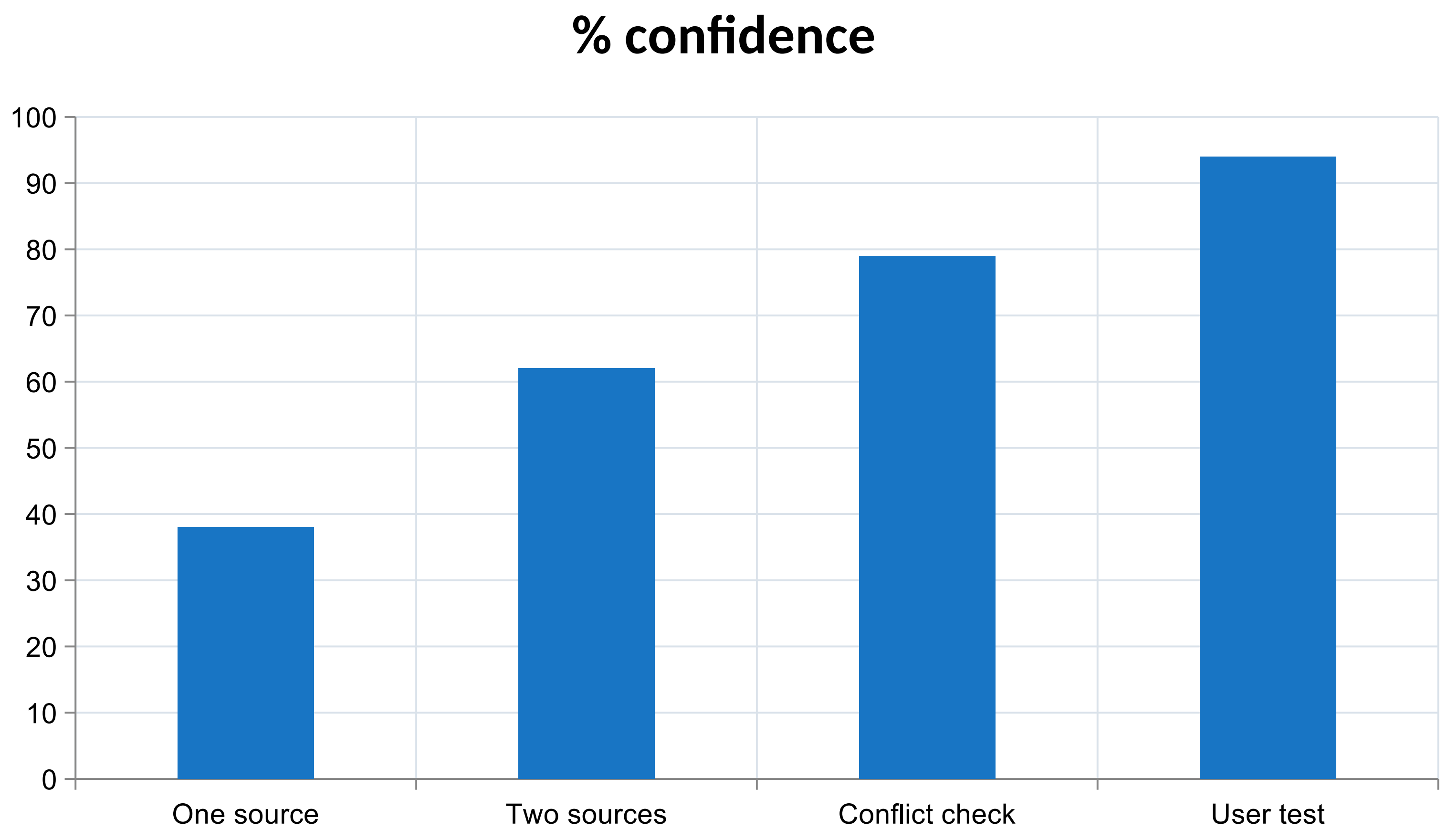
INSIGHT

Agentic AI fits work that is variable enough to need judgment but bounded enough to verify.

Illustrative synthetic values for mechanism explanation.

EDITABLE NATIVE CHART

Evidence coverage raises discovery confidence



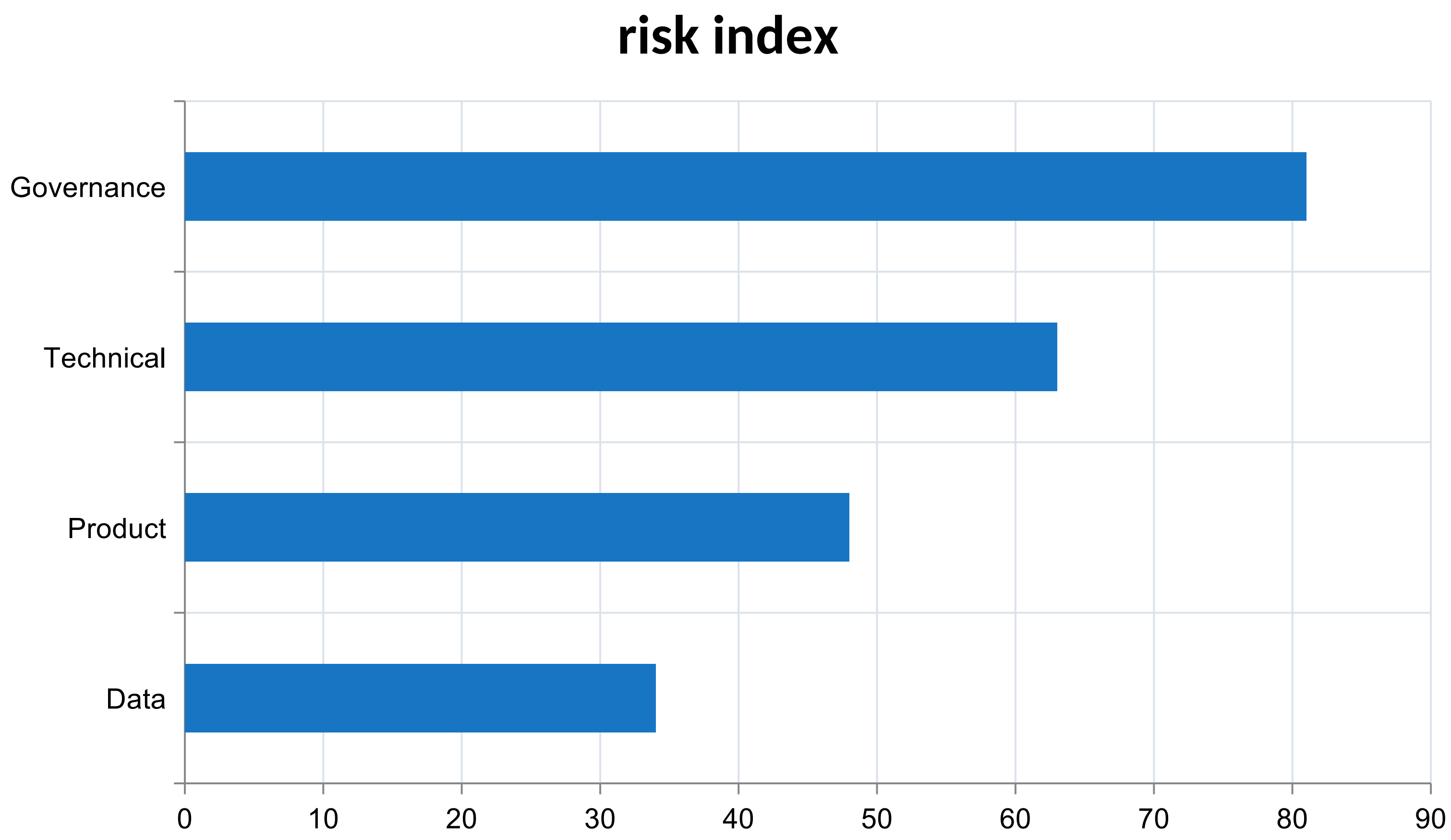
INSIGHT

Discovery confidence rises when claims survive triangulation, conflict review and user validation.

Illustrative synthetic values for mechanism explanation.

EDITABLE NATIVE CHART

Readiness gaps increase pilot exposure



INSIGHT

Builder selection depends on organisational readiness as much as product features.

Illustrative synthetic values for mechanism explanation.

Activity 01 · Frame a Product Opportunity

SCENARIO

Turn synthetic stakeholder notes into a measurable agentic-AI opportunity brief.

EXPECTED OUTPUT

Completed opportunity brief and assumption register

Detailed procedures, prompts, troubleshooting and acceptance checks are in the Learner Guide and activity PDF.

Self-contained folder: activities/activity-01-opportunity-brief/

Activity 01 · agent workflow contract

1. Inspect notes
2. Name user
3. Quantify friction
4. Define outcome
5. List assumptions
6. Set validation

Consequential actions require a current diff, passing checks and a named decision.

HUMAN GATE

Approval resumes only when the reviewed diff and current state still match.

Activity 01 · evidence and failure gates

PASS EVIDENCE

Every claim traces to a source row; the outcome is measurable; high-risk assumptions have named tests.

FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

README.pdf

detailed procedure

mock-data/

synthetic source state

templates/

learner working files

evidence/

acceptance record

Activity 02 · Compare Agentic AI Builders

SCENARIO

Run a common product-research task across three synthetic builder profiles and select a fit-for-purpose option.

EXPECTED OUTPUT

Weighted builder comparison with sensitivity check

Detailed procedures, prompts, troubleshooting and acceptance checks are in the Learner Guide and activity PDF.

Self-contained folder: activities/activity-02-builder-comparison/

WORKFLOW MAP

Activity 02 · agent workflow contract

1 Freeze task

2 Set criteria

3 Score builders

4 Apply weights

5 Test sensitivity

6 Recommend

Consequential actions require a current diff, passing checks and a named decision.

HUMAN GATE

Approval resumes only when the reviewed diff and current state still match.

Activity 02 · evidence and failure gates

PASS EVIDENCE

Scores use one common task, weights total 100%, and the recommendation survives or explains sensitivity.

FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

README.pdf

detailed procedure

mock-data/

synthetic source state

templates/

learner working files

evidence/

acceptance record

Activity 03 · Build a Research Evidence Pack

SCENARIO

Transform mock market and competitor records into a traceable, uncertainty-aware insight brief.

EXPECTED OUTPUT

Claim ledger, competitor matrix and three prioritised insights

Detailed procedures, prompts, troubleshooting and acceptance checks are in the Learner Guide and activity PDF.

Self-contained folder: activities/activity-03-research-evidence-pack/

Activity 03 · agent workflow contract

1. Load records
2. Normalise facts
3. Triangulate claims
4. Flag conflict
5. Rank insight
6. Cite sources

Each node consumes named keys, emits a status, and leaves retrievable execution evidence.

ACCEPTANCE MODEL

Activity 03 · evidence and failure gates

PASS EVIDENCE

Material insights cite at least two records or explicitly state single-source uncertainty.

FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

README.pdf

detailed procedure

mock-data/

synthetic source state

templates/

learner working files

evidence/

acceptance record

ACTIVITY BRIEF

Activity 04 · Create Persona and JTBD Map

SCENARIO

Derive evidence-based personas and jobs-to-be-done without presenting generated assumptions as facts.

EXPECTED OUTPUT

Two persona cards, JTBD statements and validation plan

Detailed procedures, prompts, troubleshooting and acceptance checks are in the Learner Guide and activity PDF.

Self-contained folder: activities/activity-04-persona-jtbd-map/

Activity 04 · agent workflow contract

1. Cluster needs
2. Draft persona
3. Write JTBD
4. Label inference
5. Rank risk
6. Plan test

Consequential actions require a current diff, passing checks and a named decision.

HUMAN GATE

Approval resumes only when the reviewed diff and current state still match.

Activity 04 · evidence and failure gates

PASS EVIDENCE

Persona needs trace to source IDs; assumptions are labelled; each risky assumption has a feasible validation test.

FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

README.pdf

detailed procedure

mock-data/

synthetic source state

templates/

learner working files

evidence/

acceptance record

CHECKPOINT

Topic 1 technical checkpoint

1. Which state field prevents a silent downstream assumption?
2. What evidence proves the mechanism completed successfully?
3. Which failure must stop rather than retry?
4. Who owns the consequential decision?

CHECKPOINT

Topic 1 transfer challenge

1. **Change one input constraint and predict the affected nodes.**
2. **Name the identifier that preserves provenance across the change.**
3. **Define one machine threshold and one human decision.**
4. **Describe the minimum evidence required to resume the run.**

2

TOPIC 2

Building and Optimising Agentic Product Development Workflows

Translate product intent into bounded agent roles, specifications, hand-offs, evaluation sets and an iterative prototype workflow that improves with evidence.

1

CONTROL

2

EVIDENCE

3

AUTOMATION

4

DECISION

Every product decision ends in observable evidence—not an agent's assertion.

01 · Goal-plan-act-observe loop: execution path

1. Interpret goal
2. Choose next step
3. Act through tool
4. Observe state

STATE PASSED BETWEEN STAGES

goal	next_action
tool_result	status

FAILURE MODE

The agent loops without progress or stops on a plausible narrative.

01 · Goal-plan-act-observe loop: contract and control

INPUT CONTRACT

required keys + owner

DETERMINISTIC GATE

schema + threshold

ACCEPTED OUTPUT

evidence + status

BLOCK / REPAIR / ESCALATE

Use explicit budgets, stop conditions and observable state transitions.

01 · Goal-plan-act-observe loop: evidence and decision

MEASURE

% loops ending in a verified stop state

ACCEPTANCE EVIDENCE

Loop trace with terminal status

OBSERVED	THRESHOLD	DECISION	OWNER
machine value	approved rule	PASS / REPAIR	named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

02 · Product context packet: execution path

1. State vision
2. Name users
3. Add constraints
4. Define evidence

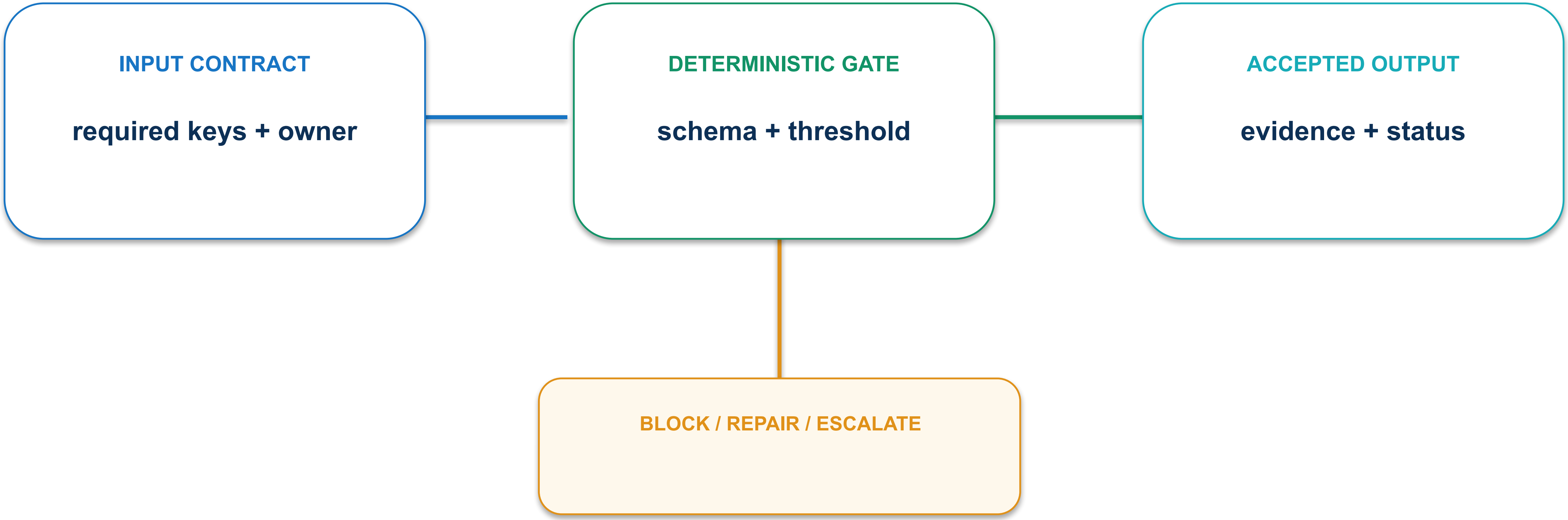
STATE PASSED BETWEEN STAGES

vision	users
constraints	acceptance

FAILURE MODE

The agent fills missing business context with generic assumptions.

02 · Product context packet: contract and control



Version a compact context packet and expose unknowns explicitly.

02 · Product context packet: evidence and decision

MEASURE

Required context fields present before execution

ACCEPTANCE EVIDENCE

Context file and assumption log

OBSERVED	THRESHOLD	DECISION	OWNER
machine value	approved rule	PASS / REPAIR	named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

03 · Outcome-focused specification: execution path

1 Describe problem

2 Define behavior

3 Set boundaries

4 Specify done

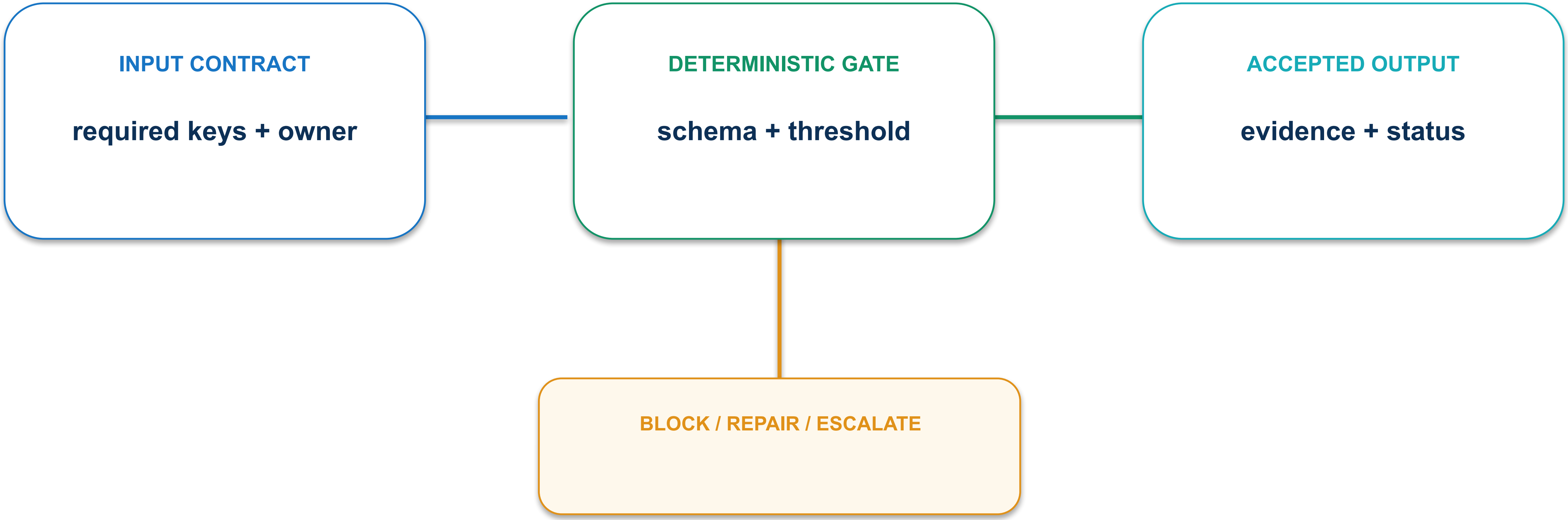
STATE PASSED BETWEEN STAGES

problem	**behavior**
non_goals	**acceptance**

FAILURE MODE

The specification dictates implementation while leaving success ambiguous.

03 · Outcome-focused specification: contract and control



Describe what must be true and what must not change.

03 · Outcome-focused specification: evidence and decision

MEASURE

% acceptance criteria observable by reviewer

ACCEPTANCE EVIDENCE

PRD section with testable criteria

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

MECHANISM

04 · User stories and acceptance criteria: execution path

1. **Name actor**
2. **State intent**
3. **Explain value**
4. **Write examples**

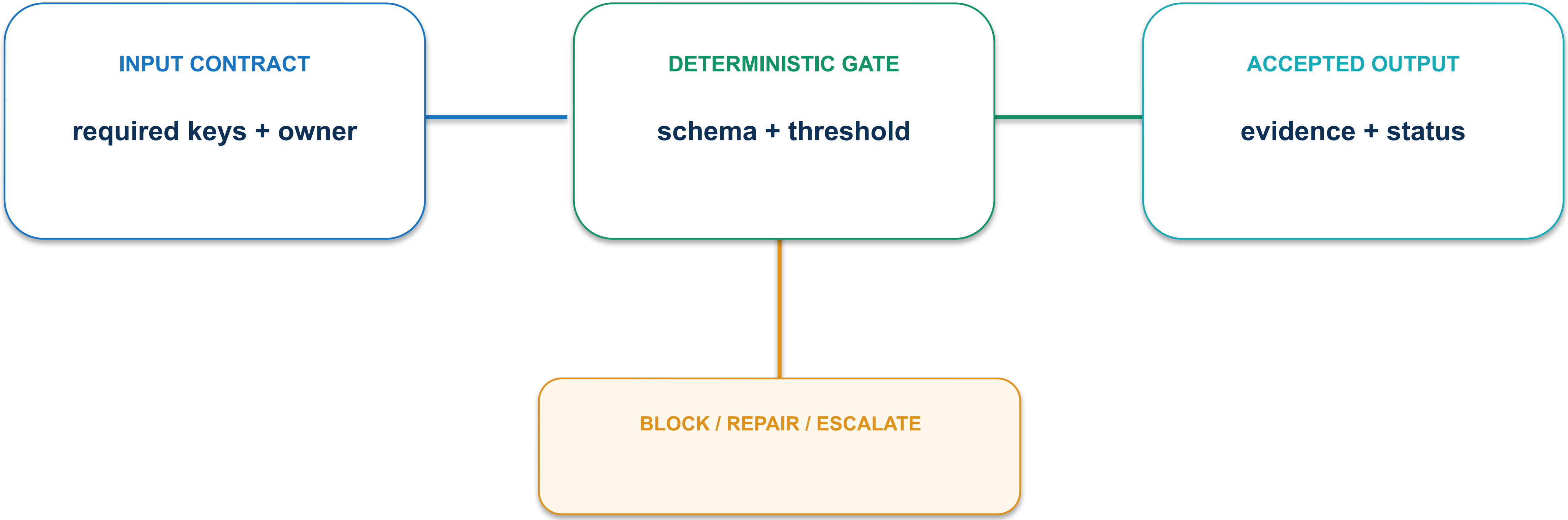
STATE PASSED BETWEEN STAGES

actor	**intent**
value	**examples**

FAILURE MODE

A story is marked complete when only the happy path works.

04 · User stories and acceptance criteria: contract and control



Attach concrete examples and negative conditions to each story.

04 · User stories and acceptance criteria: evidence and decision

MEASURE

% stories with positive, edge and failure examples

ACCEPTANCE EVIDENCE

Story map with example-based acceptance

OBSERVED	THRESHOLD	DECISION	OWNER
machine value	approved rule	PASS / REPAIR	named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

MECHANISM

05 · Journey-map orchestration: execution path

1. **Map stages**
2. **Locate friction**
3. **Assign agent role**
4. **Keep human gate**

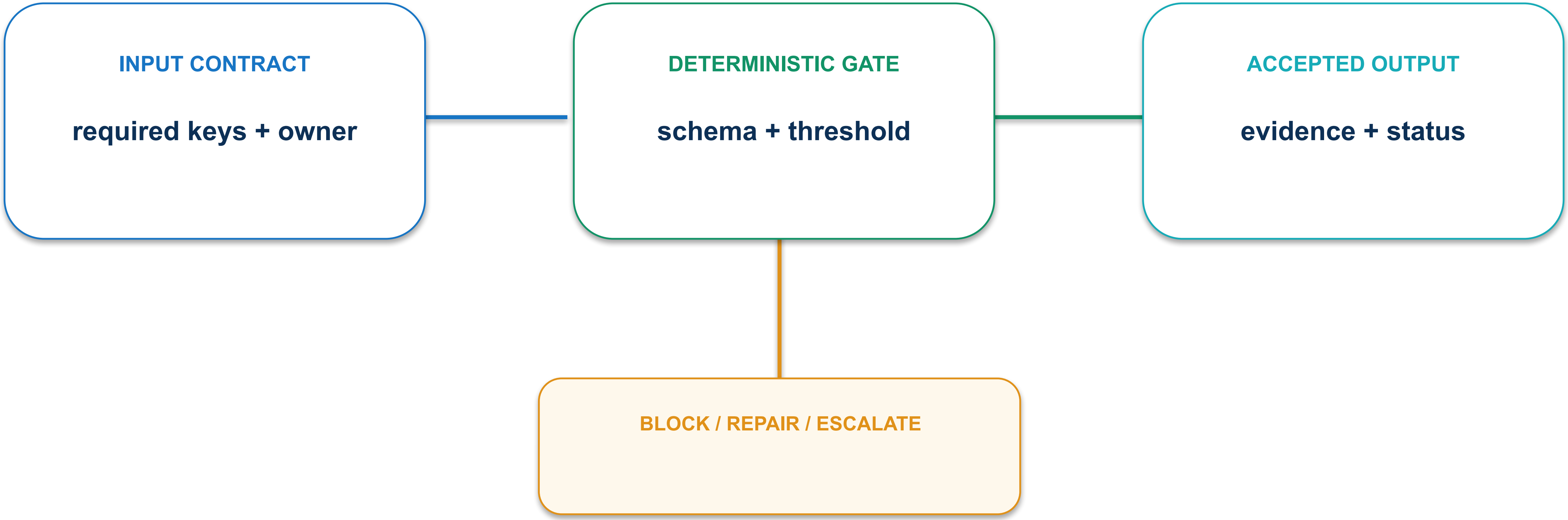
STATE PASSED BETWEEN STAGES

stage	friction
agent_role	gate

FAILURE MODE

The workflow optimises speed while weakening trust at critical moments.

05 · Journey-map orchestration: contract and control



Preserve recovery, explanation and control at consequential journey points.

05 · Journey-map orchestration: evidence and decision

MEASURE

% high-consequence moments with human control

ACCEPTANCE EVIDENCE

Journey map with agent and human swimlanes

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

06 · Single-agent baseline: execution path

1. Bound one goal
2. Connect minimal tools
3. Run baseline
4. Measure gaps

STATE PASSED BETWEEN STAGES

goal	tools
baseline	gaps

FAILURE MODE

A multi-agent design is introduced before a simpler baseline is measured.

06 · Single-agent baseline: contract and control

INPUT CONTRACT

required keys + owner

DETERMINISTIC GATE

schema + threshold

ACCEPTED OUTPUT

evidence + status

BLOCK / REPAIR / ESCALATE

Prove the minimum viable agent before adding roles.

06 · Single-agent baseline: evidence and decision

MEASURE

Accepted tasks per 20-case evaluation set

ACCEPTANCE EVIDENCE

Baseline run log and failure summary

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

MECHANISM

07 · Multi-agent role design: execution path

1. Decompose domains
2. Assign role
3. Limit authority
4. Define coordinator

STATE PASSED BETWEEN STAGES

domain	role
authority	coordinator

FAILURE MODE

Multiple agents duplicate work and amplify inconsistent assumptions.

07 · Multi-agent role design: contract and control

INPUT CONTRACT

required keys + owner

DETERMINISTIC GATE

schema + threshold

ACCEPTED OUTPUT

evidence + status

BLOCK / REPAIR / ESCALATE

Give each role distinct inputs, output schema and decision rights.

07 · Multi-agent role design: evidence and decision

MEASURE

Handoffs that add unique value / total handoffs

ACCEPTANCE EVIDENCE

Role charter and responsibility matrix

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

08 · Handoff contract: execution path

1 Validate input

2 Perform task

3 Emit schema

4 Acknowledge receipt

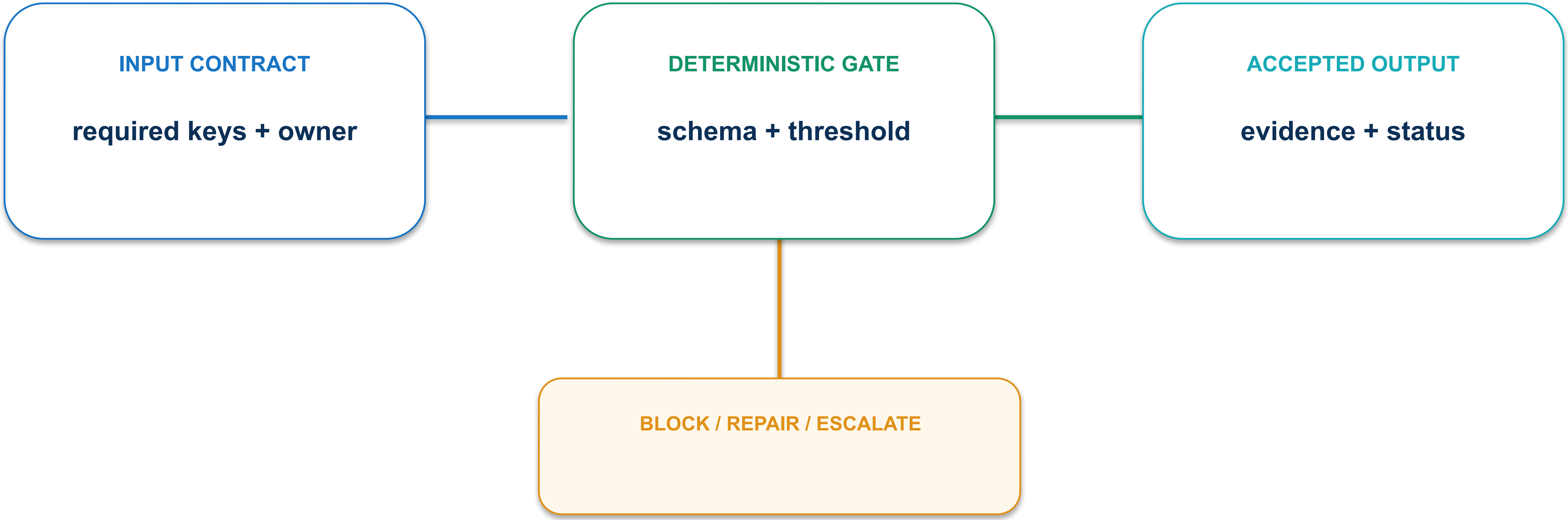
STATE PASSED BETWEEN STAGES

input_schema	task
output_schema	ack

FAILURE MODE

Free-form outputs break downstream steps or lose provenance.

08 · Handoff contract: contract and control



Use stable identifiers, structured outputs and fail-closed validation.

08 · Handoff contract: evidence and decision

MEASURE

% handoffs passing schema validation

ACCEPTANCE EVIDENCE

Handoff record with validation status

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

MECHANISM

09 · Tool and retrieval grounding: execution path

1 **Identify need**

2 **Select source**

3 **Retrieve context**

4 **Cite evidence**

STATE PASSED BETWEEN STAGES

need

source

context

citation

FAILURE MODE

The agent uses stale, irrelevant or untrusted material as product truth.

09 · Tool and retrieval grounding: contract and control

INPUT CONTRACT

required keys + owner

DETERMINISTIC GATE

schema + threshold

ACCEPTED OUTPUT

evidence + status

BLOCK / REPAIR / ESCALATE

Constrain sources, retain citations and test retrieval misses.

09 · Tool and retrieval grounding: evidence and decision

MEASURE

% answer claims grounded in approved sources

ACCEPTANCE EVIDENCE

Retrieved evidence linked to output claims

OBSERVED	THRESHOLD	DECISION	OWNER
machine value	approved rule	PASS / REPAIR	named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

MECHANISM

10 · Prompt contract design: execution path

1. State objective
2. Provide context
3. Set constraints
4. Define format

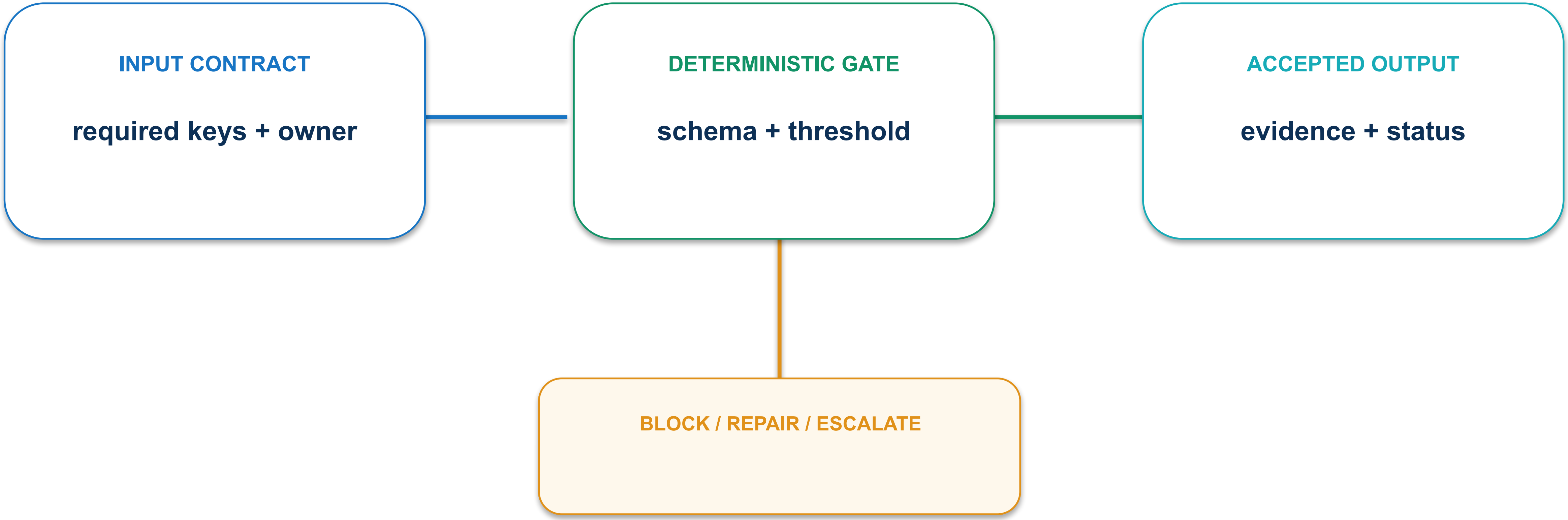
STATE PASSED BETWEEN STAGES

objective	context
constraints	output_schema

FAILURE MODE

A long prompt hides conflicting instructions and undefined priorities.

10 · Prompt contract design: contract and control



Separate goal, context, rules, examples and output contract.

10 · Prompt contract design: evidence and decision

MEASURE

First-pass outputs meeting all contract fields

ACCEPTANCE EVIDENCE

Prompt version and conformance checklist

OBSERVED	THRESHOLD	DECISION	OWNER
machine value	approved rule	PASS / REPAIR	named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

MECHANISM

11 · Evaluation-set engineering: execution path

1. Sample real tasks
2. Add edge cases
3. Freeze expected traits
4. Score runs

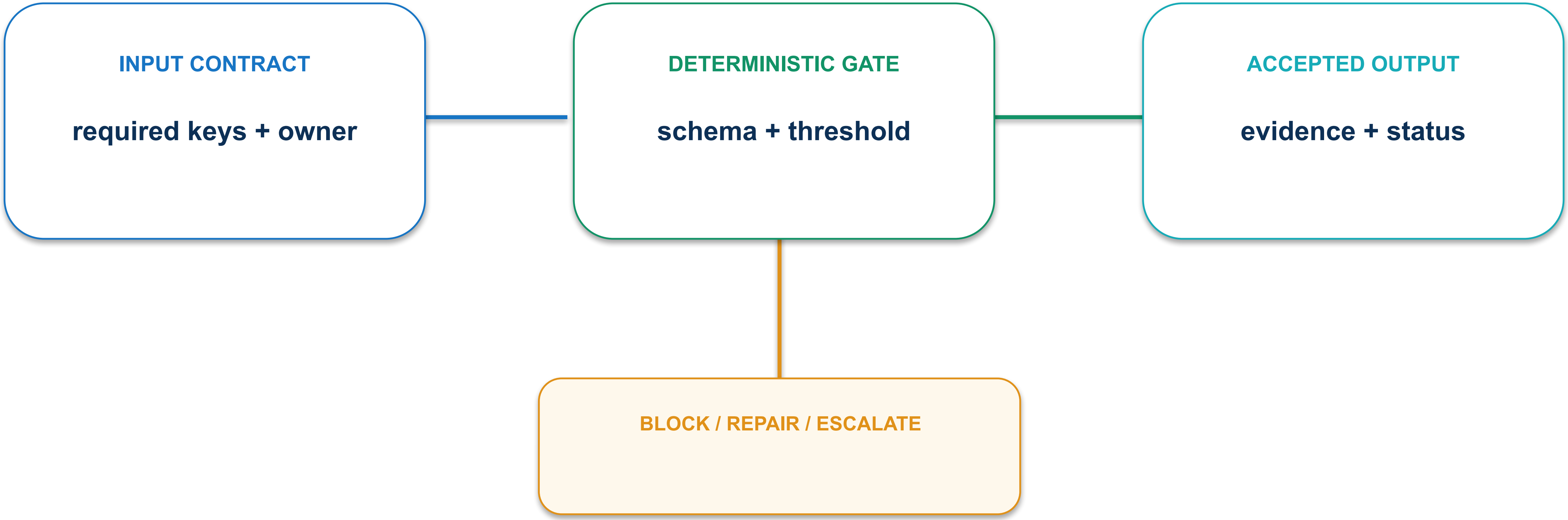
STATE PASSED BETWEEN STAGES

task_id	input
expected	score

FAILURE MODE

Teams judge prototypes from one impressive demonstration.

11 · Evaluation-set engineering: contract and control



Build a representative frozen set before optimisation.

11 · Evaluation-set engineering: evidence and decision

MEASURE

Coverage across core, edge and adversarial cases

ACCEPTANCE EVIDENCE

Versioned evaluation CSV and score report

OBSERVED	THRESHOLD	DECISION	OWNER
machine value	approved rule	PASS / REPAIR	named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

12 · Iteration and error taxonomy: execution path

1. **Classify failure**
2. **Find causal layer**
3. **Apply one change**
4. **Re-run set**

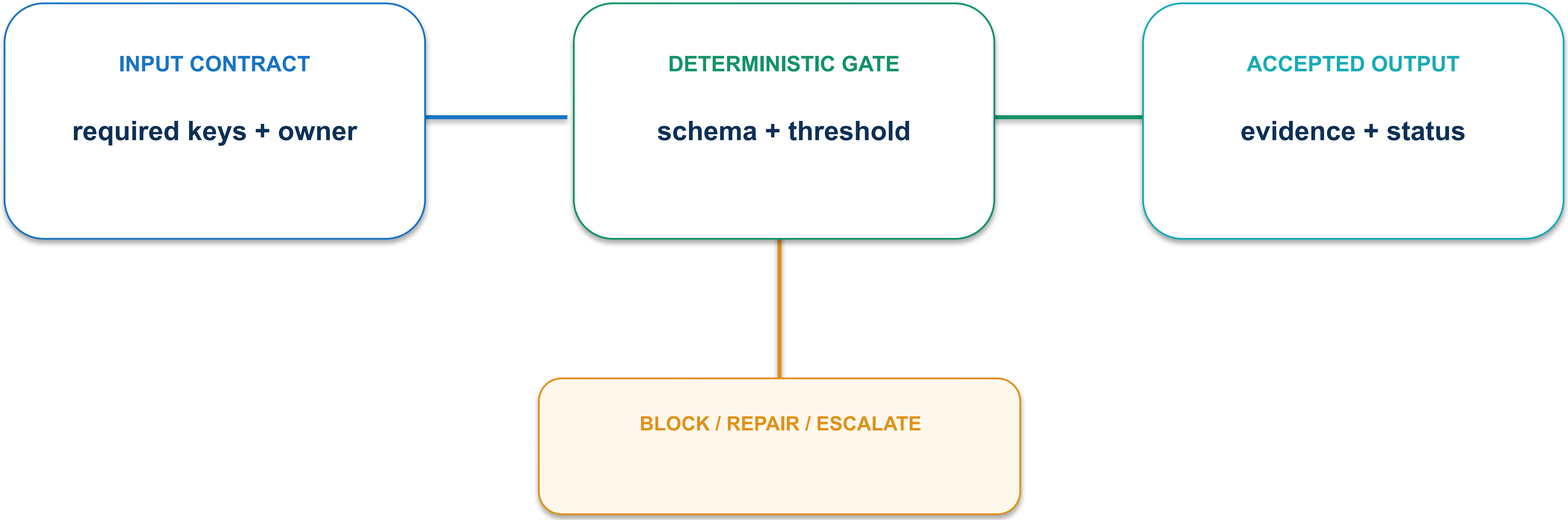
STATE PASSED BETWEEN STAGES

failure_type	**layer**
change	**delta**

FAILURE MODE

Several prompt, model and data changes are bundled, obscuring causality.

12 · Iteration and error taxonomy: contract and control



Change one causal layer per experiment and re-run the full frozen set.

12 · Iteration and error taxonomy: evidence and decision

MEASURE

Improvement on failed cases without regression

ACCEPTANCE EVIDENCE

Experiment ledger with before-after deltas

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

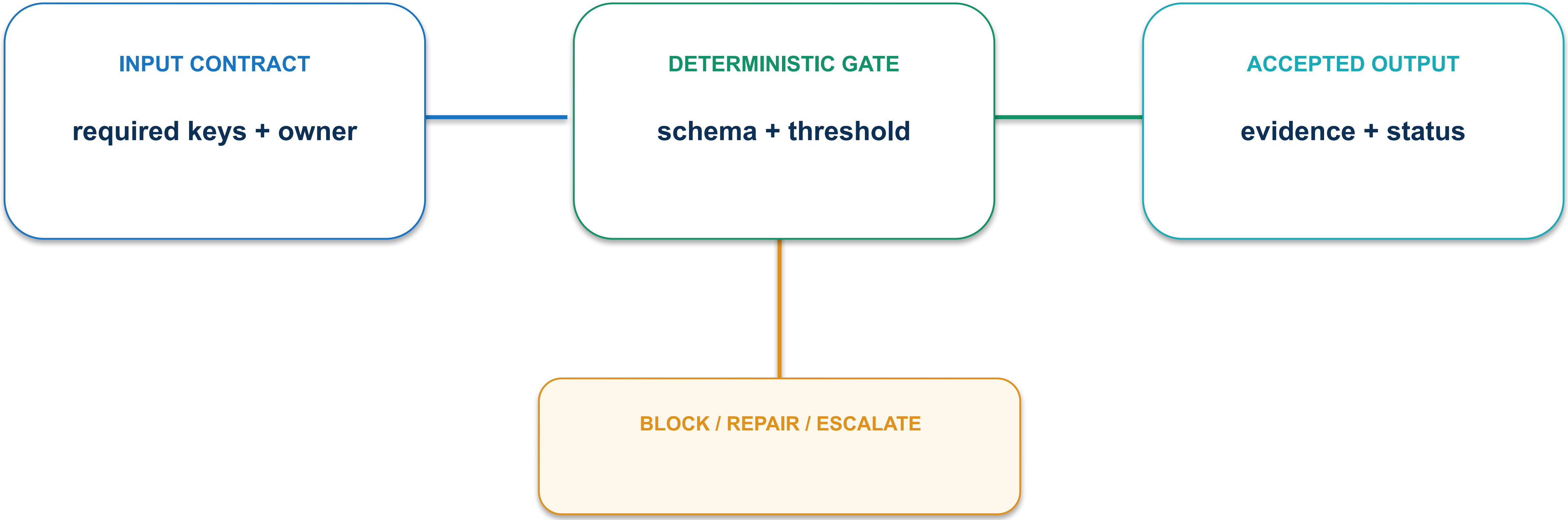
13 · Prototype-to-learning loop: execution path

PROTOTYPE LEARNING SURFACE

1	BUILD	Create the thinnest useful slice
2	TEST	Observe representative users
3	LEARN	Separate signal from opinion
4	DECIDE	Continue, pivot or stop

CONCEPT VISUAL • LABELLED

13 · Prototype-to-learning loop: contract and control



Time-box builds and define the decision each prototype must unlock.

13 · Prototype-to-learning loop: evidence and decision

MEASURE

Validated assumptions per prototype cycle

ACCEPTANCE EVIDENCE

Experiment brief and go-pivot-stop decision

OBSERVED	THRESHOLD	DECISION	OWNER
machine value	approved rule	PASS / REPAIR	named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

Workflow boundary: context, agent roles and human gates

1

CONTEXT

Define outcome and constraints

2

WORKFLOW

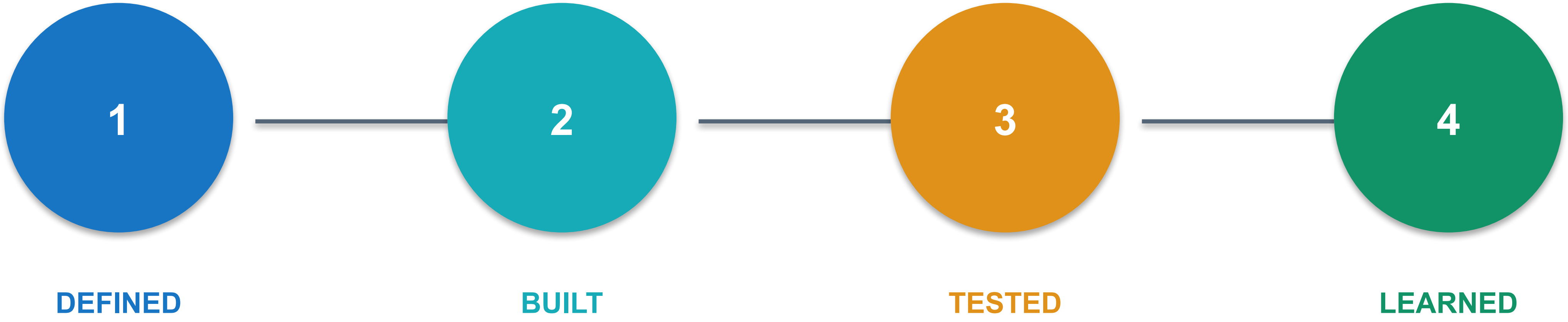
Coordinate roles and tools

3

HUMAN GATE

Review evidence and consequence

Prototype state: DEFINED, BUILT, TESTED and LEARNED



LEARNED means the prototype changed or confirmed a product hypothesis.

Product evidence chain: requirement → prototype → evaluation → decision

requirement_id
intended behavior

prototype_id
testable artifact

eval_run
observed result

decision_id
next posture

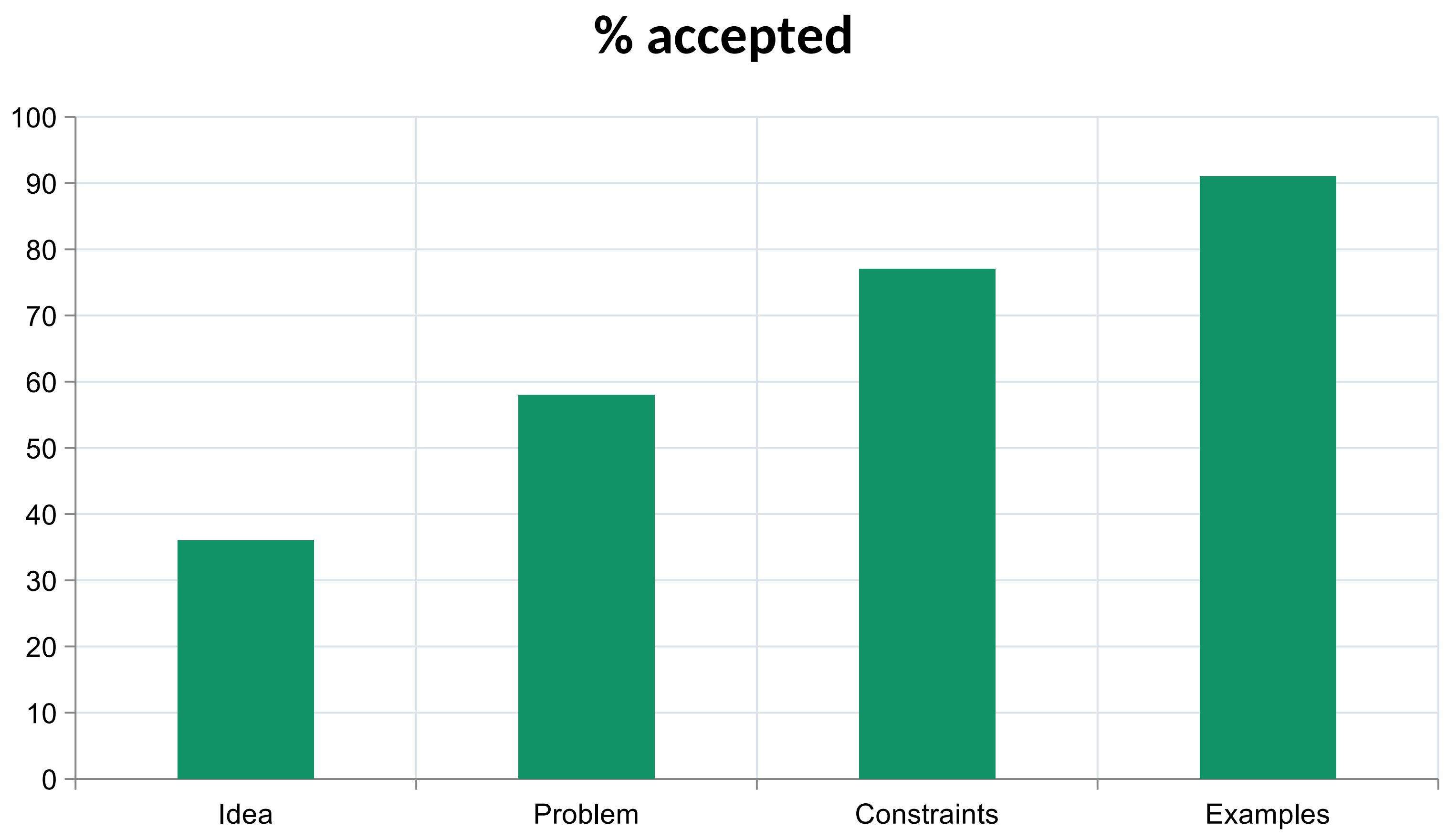
A prototype decision is credible when requirements, tests and learning remain connected.

Optimisation taxonomy: context, handoff, quality and regression

CONTEXT	repair missing product constraints
HANDOFF	restore schema and provenance
QUALITY	target the causal prompt or data layer
REGRESSION	re-run the frozen evaluation set

EDITABLE NATIVE CHART

Structured context improves first-pass quality



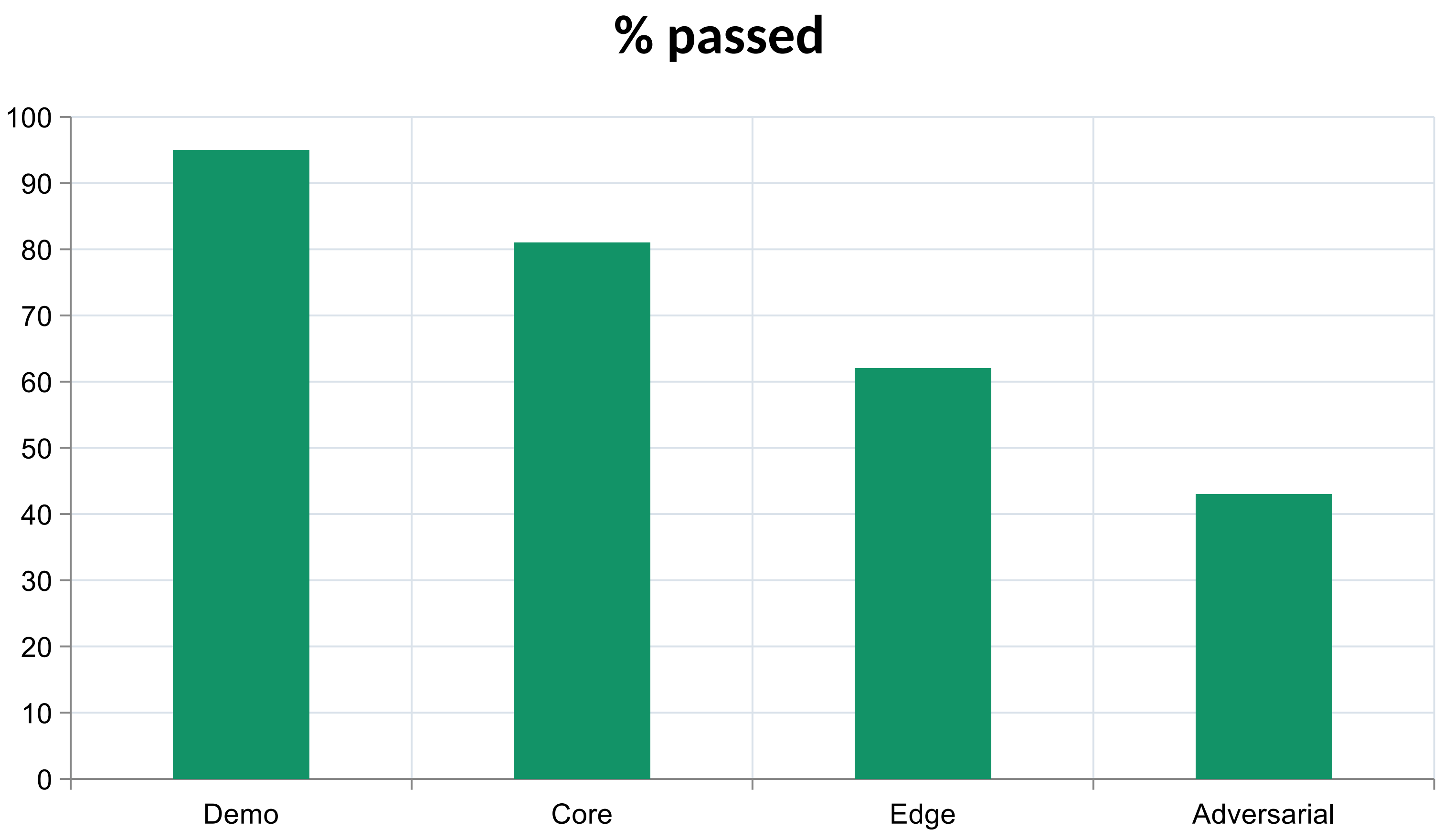
INSIGHT

Product context, constraints and examples improve first-pass usefulness.

Illustrative synthetic values for mechanism explanation.

EDITABLE NATIVE CHART

Frozen evaluations expose real improvement



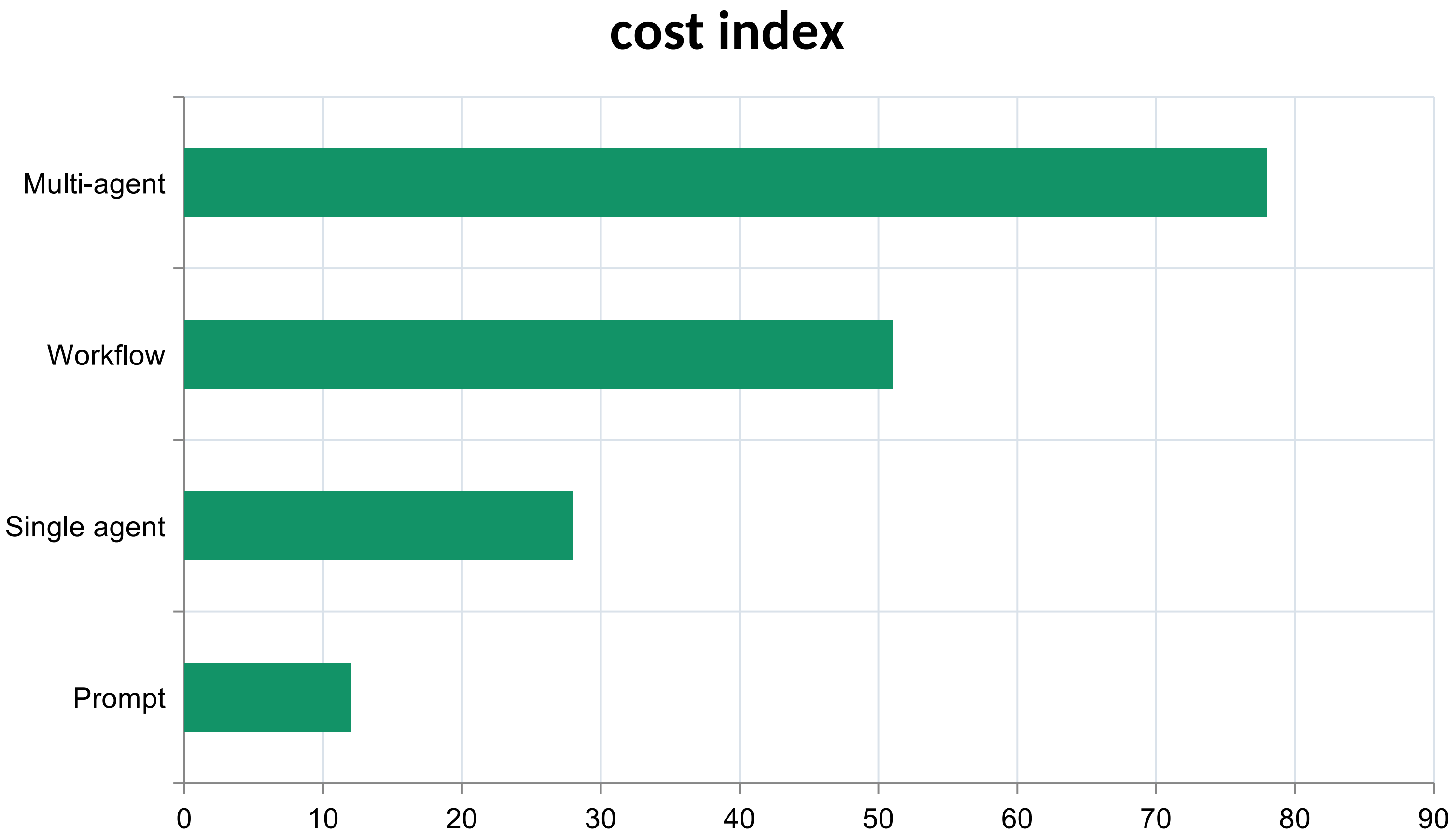
INSIGHT

A frozen evaluation set prevents teams from moving the goalposts during optimisation.

Illustrative synthetic values for mechanism explanation.

EDITABLE NATIVE CHART

Workflow complexity expands coordination cost



INSIGHT

More agent roles add coordination cost and must earn their place through measured value.

Illustrative synthetic values for mechanism explanation.

ACTIVITY BRIEF

Activity 05 · Design an Agent Workflow Canvas

SCENARIO

Map a product-discovery task into an agentic loop with tools, state, stop conditions and human gates.

EXPECTED OUTPUT

Workflow canvas and responsibility matrix

Detailed procedures, prompts, troubleshooting and acceptance checks are in the Learner Guide and activity PDF.

Self-contained folder: activities/activity-05-agent-workflow-canvas/

Activity 05 · agent workflow contract

1. Define goal
2. Map stages
3. Assign tools
4. Set schemas
5. Add gates
6. Set stop

Consequential actions require a current diff, passing checks and a named decision.

HUMAN GATE

Approval resumes only when the reviewed diff and current state still match.

Activity 05 · evidence and failure gates

PASS EVIDENCE

Every stage has an owner and observable output; consequential actions stop for human review.

FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

README.pdf

detailed procedure

mock-data/

synthetic source state

templates/

learner working files

evidence/

acceptance record

Activity 06 · Create an AI-Ready PRD

SCENARIO

Convert an opportunity brief into outcome-focused requirements, user stories and acceptance examples.

EXPECTED OUTPUT

Lean PRD, story map and acceptance set

Detailed procedures, prompts, troubleshooting and acceptance checks are in the Learner Guide and activity PDF.

Self-contained folder: activities/activity-06-prd-and-user-stories/

Activity 06 · agent workflow contract

1. Restate problem
2. Define behavior
3. Write stories
4. Add examples
5. Set non-goals
6. Review gaps

Consequential actions require a current diff, passing checks and a named decision.

HUMAN GATE

Approval resumes only when the reviewed diff and current state still match.

Activity 06 · evidence and failure gates

PASS EVIDENCE

Requirements are testable, non-goals are explicit, and every priority story has three acceptance examples.

FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

README.pdf

detailed procedure

mock-data/

synthetic source state

templates/

learner working files

evidence/

acceptance record

Activity 07 · Engineer a Prompt and Evaluation Set

SCENARIO

Create a structured prompt contract and a frozen evaluation set for a product-feedback agent.

EXPECTED OUTPUT

Versioned prompt, 20-case evaluation CSV and scoring rubric

Detailed procedures, prompts, troubleshooting and acceptance checks are in the Learner Guide and activity PDF.

Self-contained folder: activities/activity-07-prompt-evaluation-set/

Activity 07 · agent workflow contract

1. Define contract
2. Sample cases
3. Add edges
4. Freeze rubric
5. Run baseline
6. Record gaps

Each node consumes named keys, emits a status, and leaves retrievable execution evidence.

Activity 07 · evidence and failure gates

PASS EVIDENCE

The evaluation set covers all segments; scoring rules are reproducible; baseline failures are retained.

FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

README.pdf

detailed procedure

mock-data/

synthetic source state

templates/

learner working files

evidence/

acceptance record

ACTIVITY BRIEF

Activity 08 · Run a Prototype Experiment

SCENARIO

Evaluate a synthetic agent prototype, classify failures and recommend one causal improvement.

EXPECTED OUTPUT

Experiment ledger with before-after scores and decision

Detailed procedures, prompts, troubleshooting and acceptance checks are in the Learner Guide and activity PDF.

Self-contained folder: activities/activity-08-prototype-experiment/

WORKFLOW MAP

Activity 08 · agent workflow contract

1. Run baseline
2. Score outputs
3. Classify failure
4. Change one layer
5. Re-run set
6. Decide

Consequential actions require a current diff, passing checks and a named decision.

HUMAN GATE

Approval resumes only when the reviewed diff and current state still match.

Activity 08 · evidence and failure gates

PASS EVIDENCE

The change targets one causal layer, the frozen set is preserved, and regression risk is documented.

FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

README.pdf

detailed procedure

mock-data/

synthetic source state

templates/

learner working files

evidence/

acceptance record

CHECKPOINT

Topic 2 technical checkpoint

1. Which state field prevents a silent downstream assumption?

2. What evidence proves the mechanism completed successfully?

3. Which failure must stop rather than retry?

4. Who owns the consequential decision?

Topic 2 transfer challenge

1. Change one input constraint and predict the affected nodes.

2. Name the identifier that preserves provenance across the change.

3. Define one machine threshold and one human decision.

4. Describe the minimum evidence required to resume the run.

3

TOPIC 3

Deploying and Evaluating Agentic AI-Powered Product Solutions

Design a deployment posture that preserves human control, security, observability, cost discipline and a production feedback loop.

1 CONTROL

2 EVIDENCE

3 AUTOMATION

4 DECISION

Every product decision ends in observable evidence—not an agent's assertion.

01 · Prototype-to-production gate: execution path

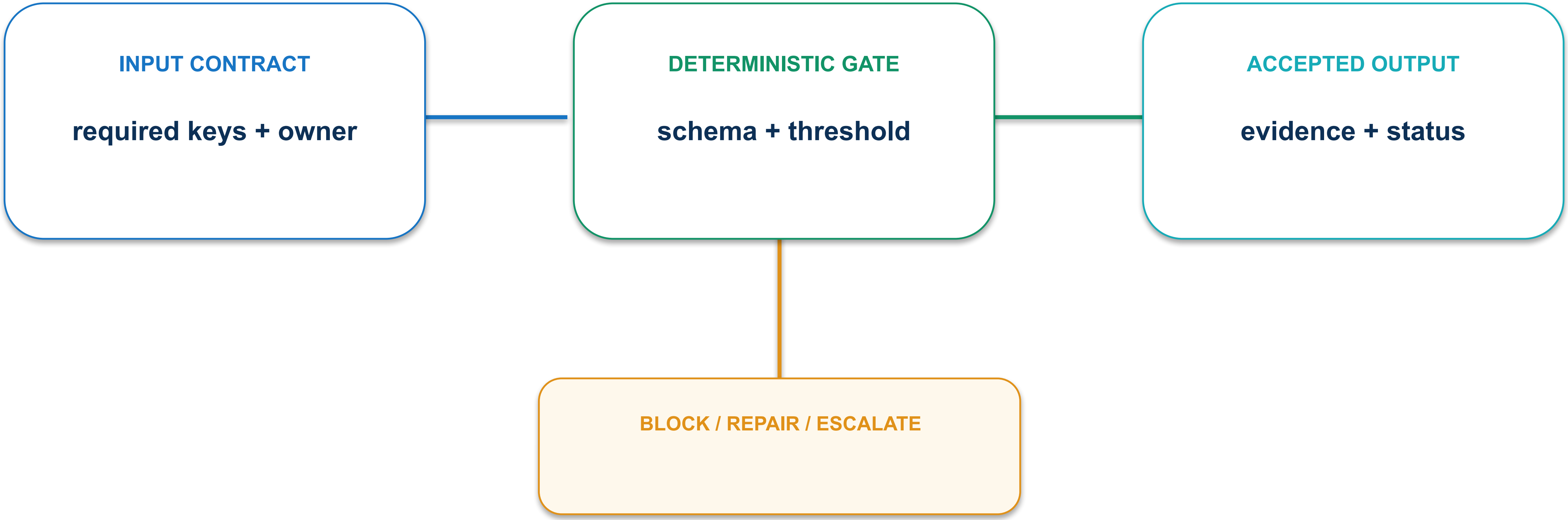
PRODUCTION READINESS REVIEW

1	VALUE	Validate user and business outcome
2	QUALITY	Pass frozen and live evaluations
3	OPERATE	Instrument cost, latency and risk
4	GOVERN	Bind rollout to human ownership

CONCEPT VISUAL • LABELLED

CONTROL SURFACE

01 · Prototype-to-production gate: contract and control



Separate learning code from supported production architecture and evidence.

01 · Prototype-to-production gate: evidence and decision

MEASURE

Required gates passed before production investment

ACCEPTANCE EVIDENCE

Handoff pack with eval set, edge cases and prompt strategy

OBSERVED	THRESHOLD	DECISION	OWNER
machine value	approved rule	PASS / REPAIR	named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

02 · Deployment channel fit: execution path

1 Map user context

2 Assess channel limits

3 Design fallback

4 Select channel

STATE PASSED BETWEEN STAGES

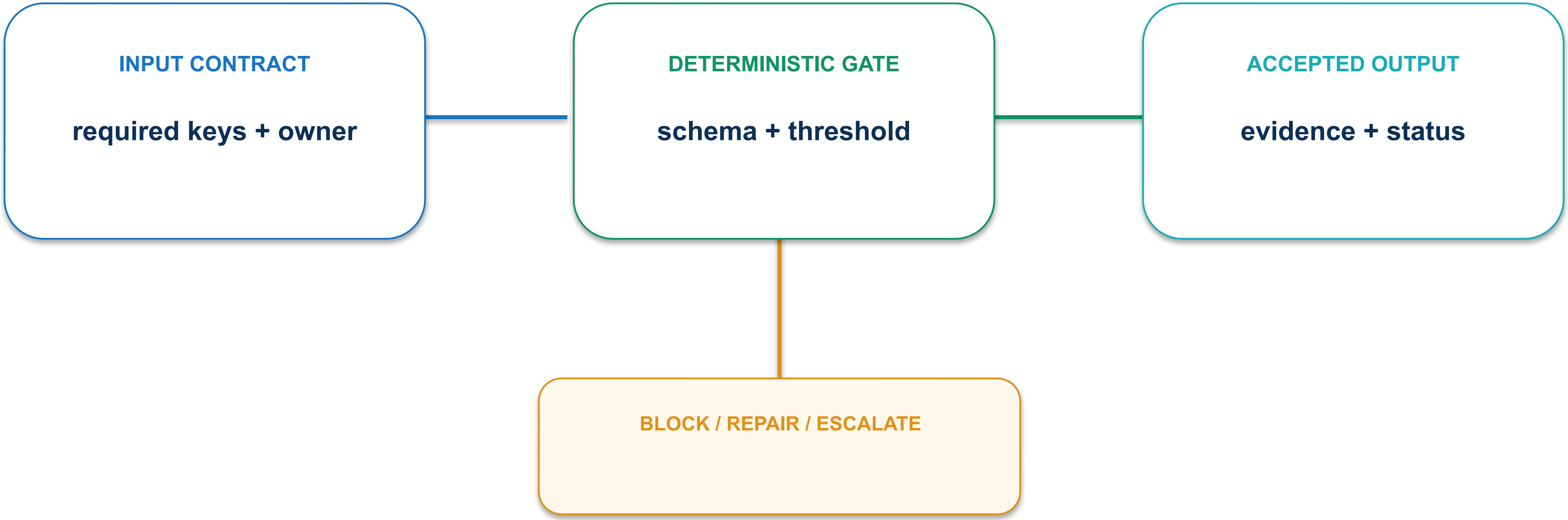
context	channel
fallback	choice

FAILURE MODE

The easiest integration is chosen despite poor user context or control.

CONTROL SURFACE

02 · Deployment channel fit: contract and control



Compare web, workspace, messaging and embedded channels on task and risk.

02 · Deployment channel fit: evidence and decision

MEASURE

Task completion and recovery rate by channel

ACCEPTANCE EVIDENCE

Channel matrix and fallback path

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

MECHANISM

03 · Reference deployment architecture: execution path

1. **Receive request**
2. **Orchestrate agent**
3. **Access tools**
4. **Record outcome**

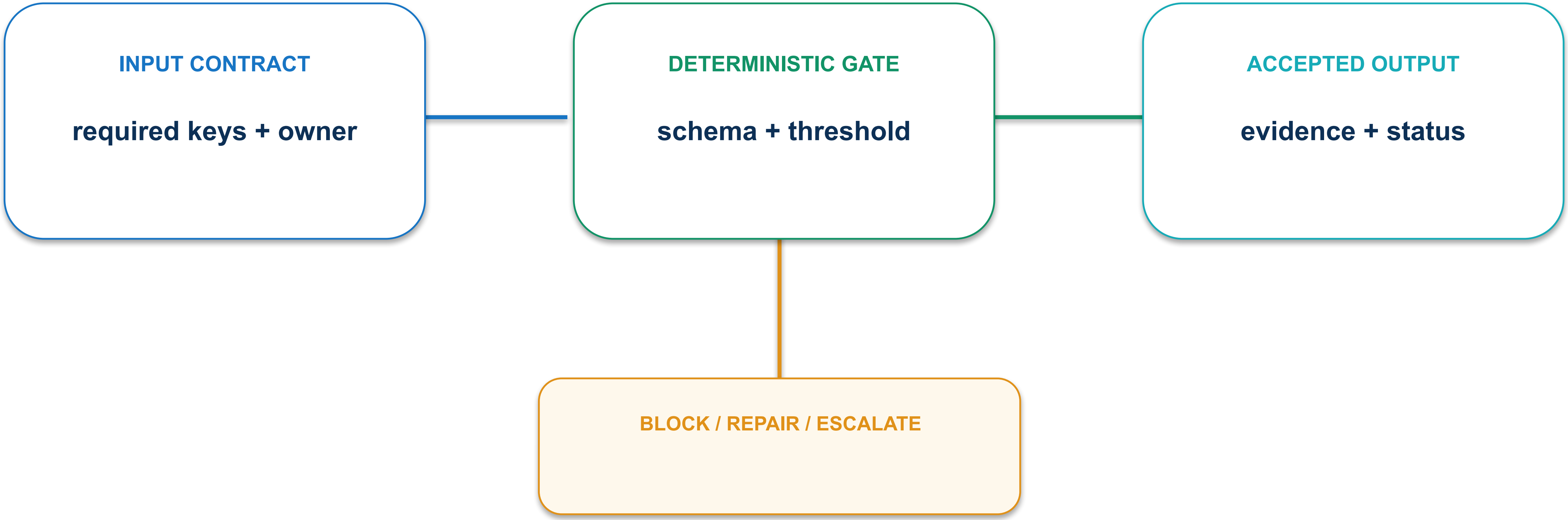
STATE PASSED BETWEEN STAGES

request_id	**orchestrator**
tool_policy	**audit_event**

FAILURE MODE

Model calls, tool actions and user decisions cannot be correlated.

03 · Reference deployment architecture: contract and control



Carry a stable request identifier through every component.

03 · Reference deployment architecture: evidence and decision

MEASURE

% transactions traceable end to end

ACCEPTANCE EVIDENCE

Architecture diagram and correlated trace

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

04 · API and integration contract: execution path

1. **Authenticate caller**
2. **Validate request**
3. **Execute bounded action**
4. **Return status**

STATE PASSED BETWEEN STAGES

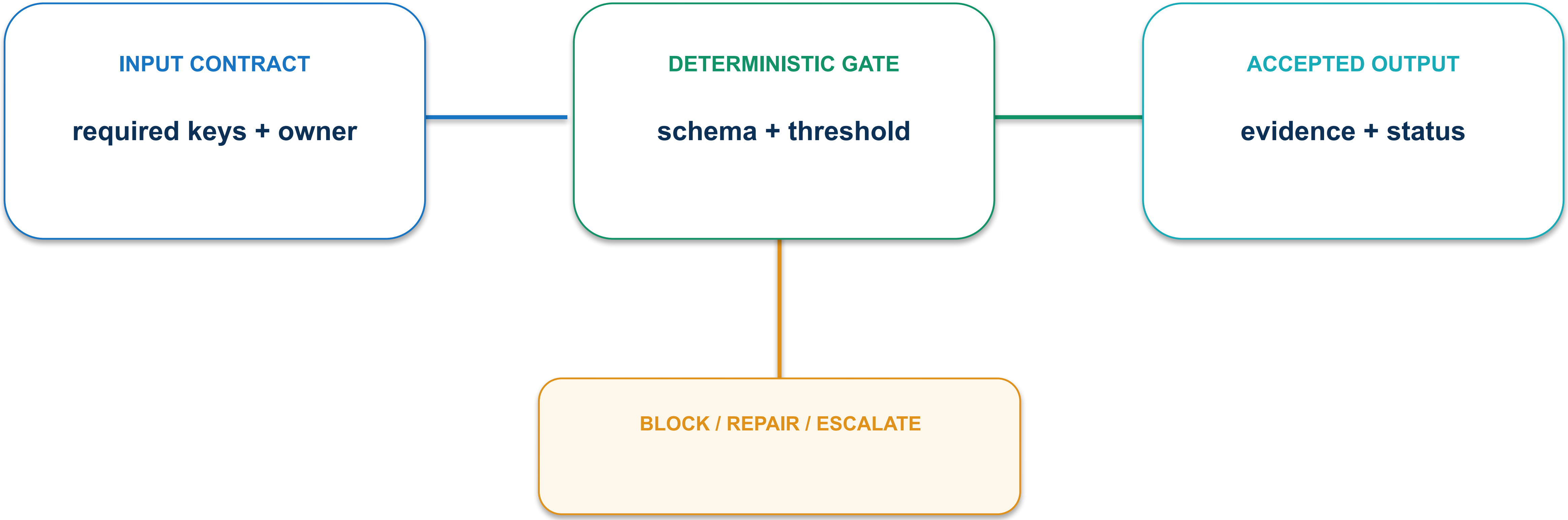
identity	**schema**
action	**status**

FAILURE MODE

Loose integrations permit malformed or over-broad actions.

CONTROL SURFACE

04 · API and integration contract: contract and control



Use typed schemas, idempotency and explicit error states.

04 · API and integration contract: evidence and decision

MEASURE

% requests with schema-valid input and explicit outcome

ACCEPTANCE EVIDENCE

API contract and negative-test results

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

05 · Programming skill mix: execution path

1. **Specify behavior**
2. **Read interfaces**
3. **Test integration**
4. **Operate service**

STATE PASSED BETWEEN STAGES

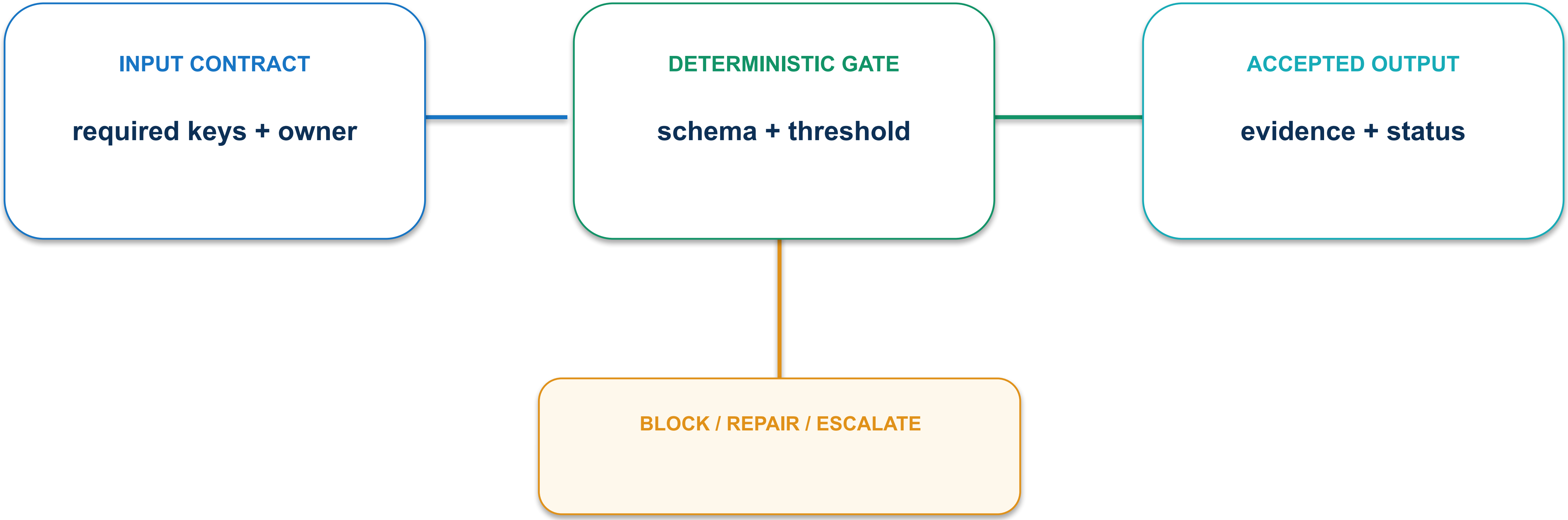
prompting | **api**

testing | **operations**

FAILURE MODE

Natural-language prototyping is assumed to replace integration and operational expertise.

05 · Programming skill mix: contract and control



Plan the product, prompt, data, integration, testing and operations skills explicitly.

05 · Programming skill mix: evidence and decision

MEASURE

Critical skill roles assigned with accountable owners

ACCEPTANCE EVIDENCE

Skills matrix and gap plan

OBSERVED	THRESHOLD	DECISION	OWNER
machine value	approved rule	PASS / REPAIR	named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

06 · Security and privacy boundary: execution path

1. **Classify data**
2. **Minimise access**
3. **Protect tools**
4. **Audit use**

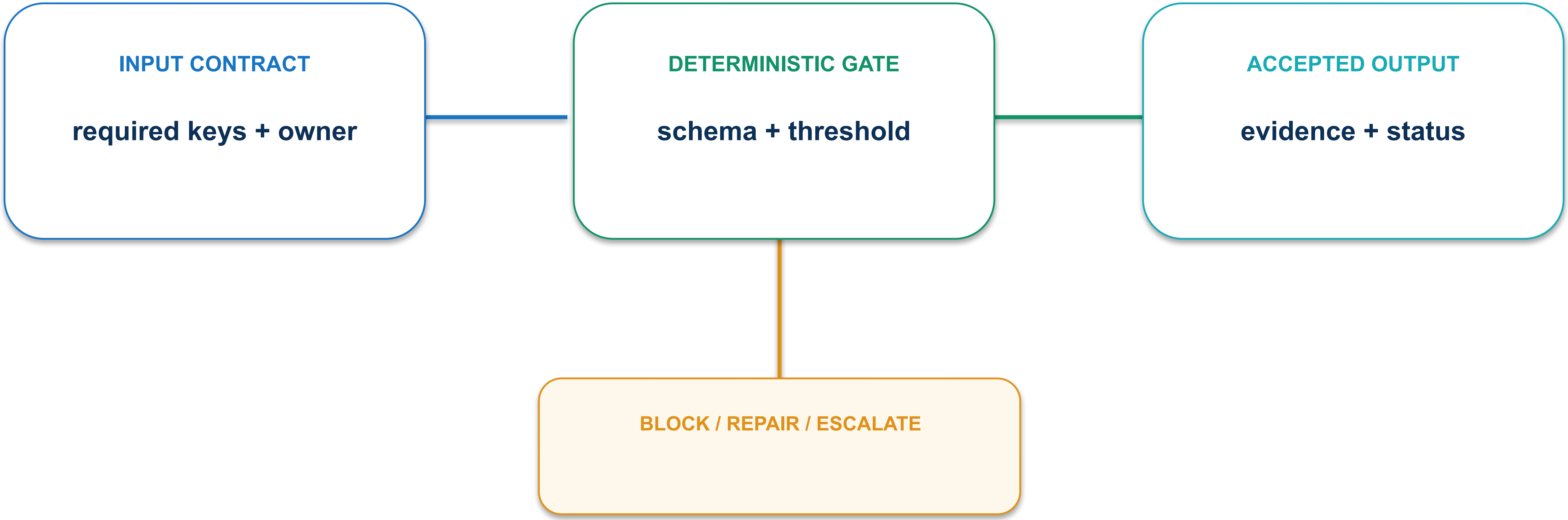
STATE PASSED BETWEEN STAGES

data_class	**permission**
tool_policy	**audit**

FAILURE MODE

Prompt injection or excess permissions turn untrusted content into consequential actions.

06 · Security and privacy boundary: contract and control



Layer input controls, tool allowlists, approval and monitoring.

06 · Security and privacy boundary: evidence and decision

MEASURE

% sensitive paths protected by least privilege

ACCEPTANCE EVIDENCE

Threat model, permission map and tested controls

OBSERVED	THRESHOLD	DECISION	OWNER
machine value	approved rule	PASS / REPAIR	named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

MECHANISM

07 · Human oversight design: execution path

1. **Classify consequence**
2. **Set confidence gate**
3. **Route exception**
4. **Record decision**

STATE PASSED BETWEEN STAGES

risk_class	**threshold**
reviewer	**decision**

FAILURE MODE

A nominal approval step becomes routine click-through.

07 · Human oversight design: contract and control

INPUT CONTRACT

required keys + owner

DETERMINISTIC GATE

schema + threshold

ACCEPTED OUTPUT

evidence + status

BLOCK / REPAIR / ESCALATE

Show the evidence and consequence at the point of decision.

07 · Human oversight design: evidence and decision

MEASURE

% consequential actions with meaningful human review

ACCEPTANCE EVIDENCE

Approval log tied to current output

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

MECHANISM

08 · Observability and product KPIs: execution path

1. **Instrument event**
2. **Compute metric**
3. **Segment failure**
4. **Trigger action**

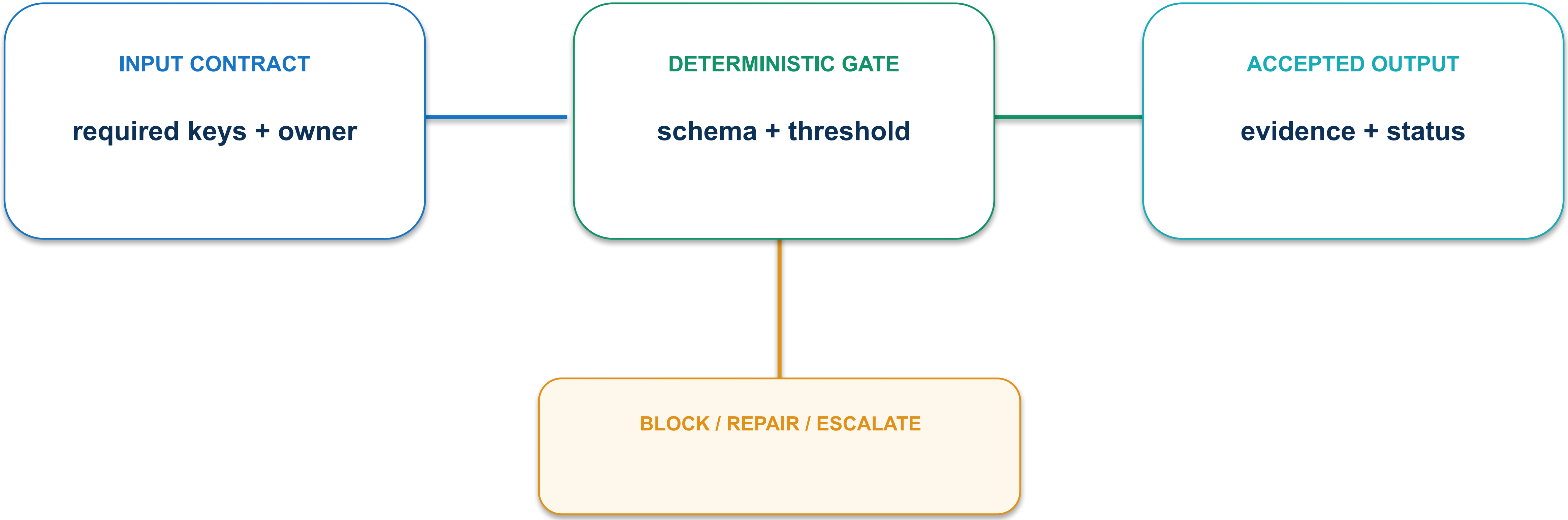
STATE PASSED BETWEEN STAGES

event	**metric**
segment	**action**

FAILURE MODE

Usage growth masks falling answer quality or rising correction work.

08 · Observability and product KPIs: contract and control



Pair business outcomes with reliability, safety and experience measures.

08 · Observability and product KPIs: evidence and decision

MEASURE

North-star outcome plus quality, cost and risk guardrails

ACCEPTANCE EVIDENCE

Metric tree and alert ownership

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

09 · Quality, latency and cost frontier: execution path

1. **Measure quality**
2. **Measure latency**
3. **Load full cost**
4. **Choose frontier**

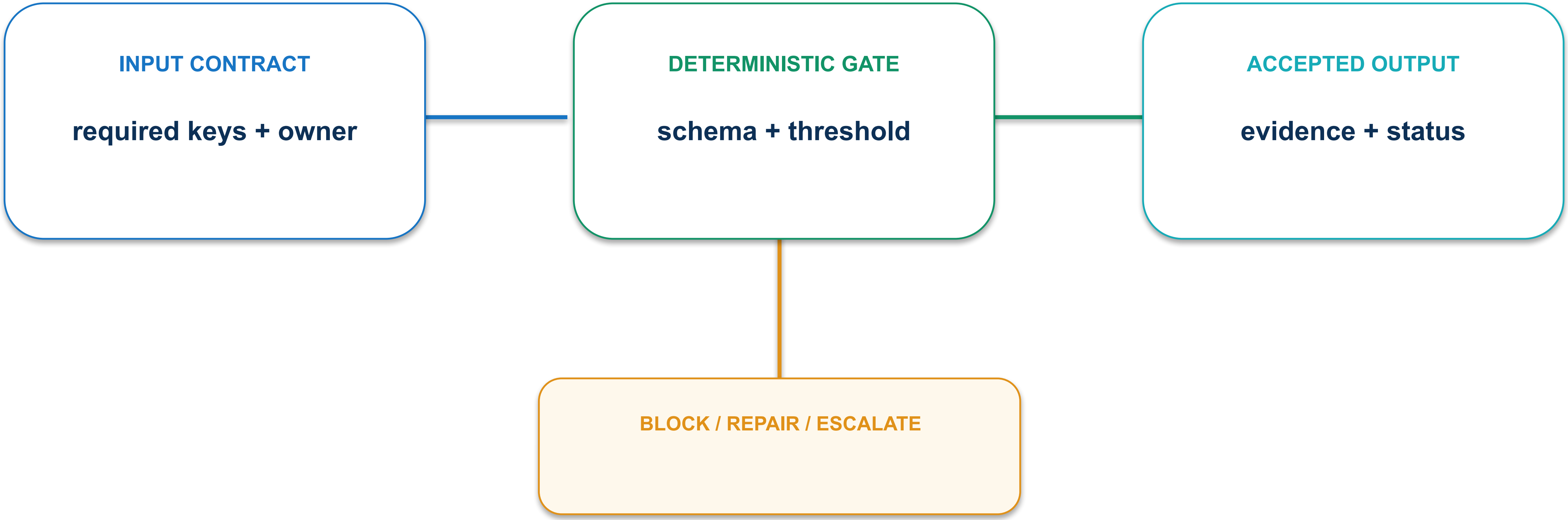
STATE PASSED BETWEEN STAGES

quality	**latency**
cost	**posture**

FAILURE MODE

The cheapest or fastest model is chosen without quality-adjusted comparison.

09 · Quality, latency and cost frontier: contract and control



Compare only configurations meeting minimum quality and risk thresholds.

09 · Quality, latency and cost frontier: evidence and decision

MEASURE

Accepted outcome cost at target response time

ACCEPTANCE EVIDENCE

Pareto chart and selected operating point

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

MECHANISM

10 · Production evaluation loop: execution path

1. **Sample traffic**
2. **Score outcomes**
3. **Review failures**
4. **Release repair**

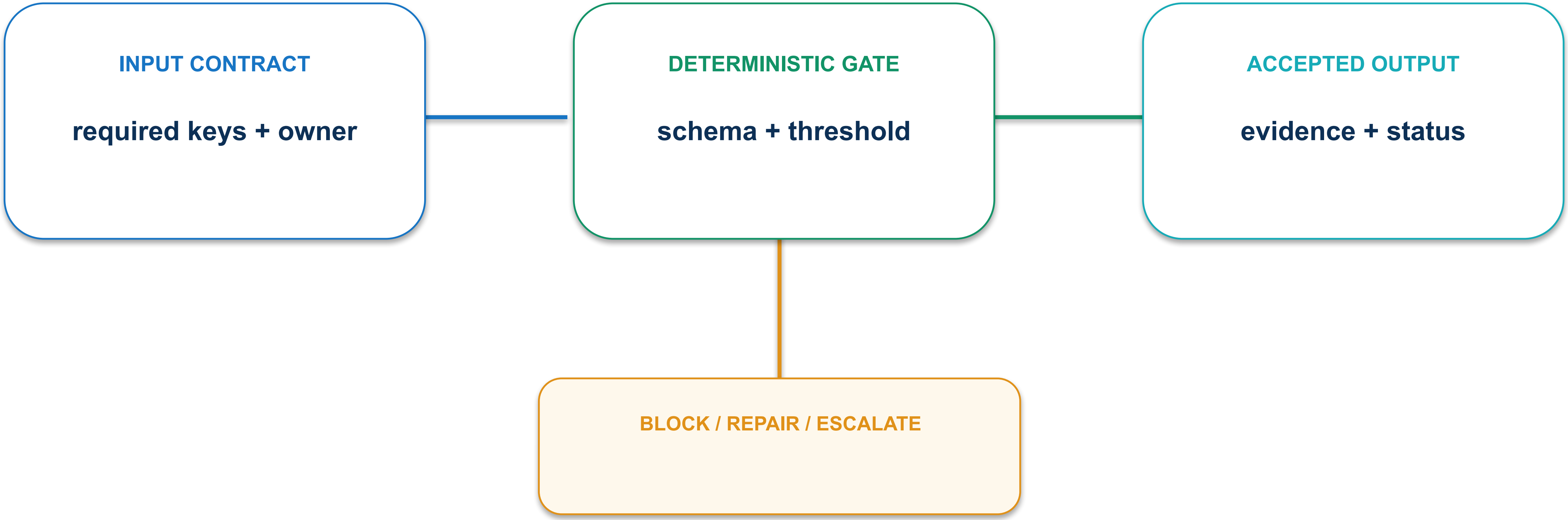
STATE PASSED BETWEEN STAGES

sample	**score**
failure	**release**

FAILURE MODE

Offline evals remain green while live inputs drift.

10 · Production evaluation loop: contract and control



Combine frozen regression sets with representative live sampling.

10 · Production evaluation loop: evidence and decision

MEASURE

% production samples scored and actioned

ACCEPTANCE EVIDENCE

Eval dashboard and linked repair record

OBSERVED	THRESHOLD	DECISION	OWNER
machine value	approved rule	PASS / REPAIR	named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

MECHANISM

11 · Rollout and change management: execution path

1. Pilot cohort
2. Train users
3. Expand gradually
4. Retire safely

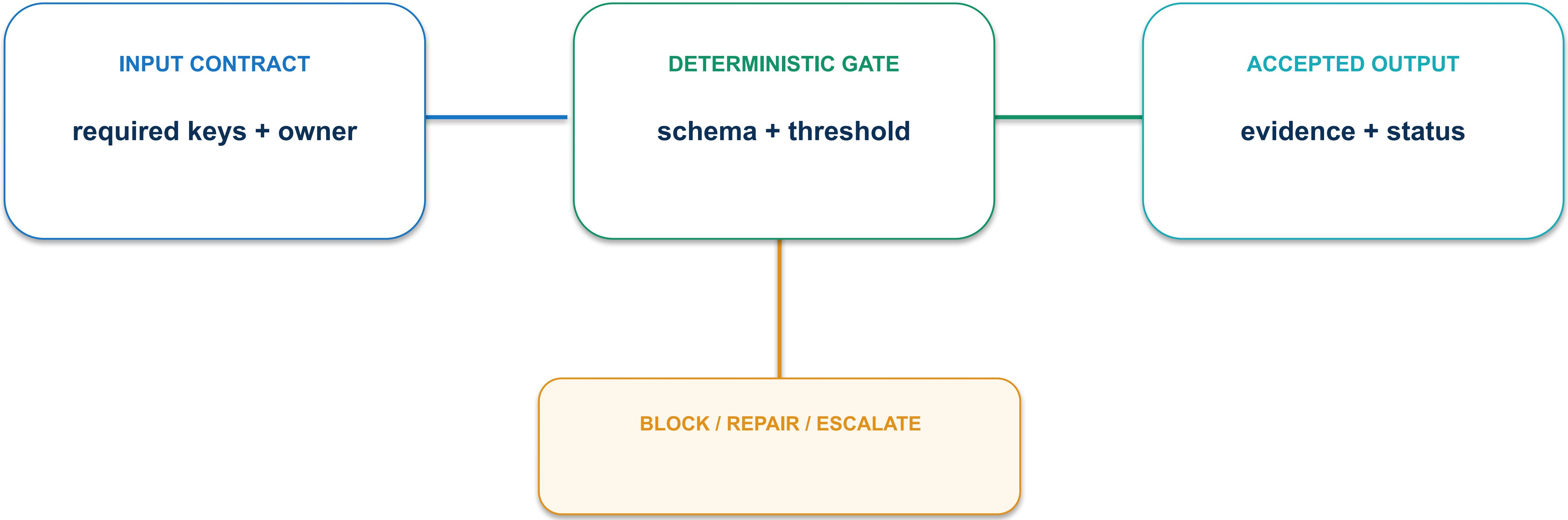
STATE PASSED BETWEEN STAGES

cohort	**enablement**
gate	**rollback**

FAILURE MODE

A big-bang launch creates unmanaged behavior and support load.

11 · Rollout and change management: contract and control



Use staged rollout, telemetry, support playbooks and rollback criteria.

11 · Rollout and change management: evidence and decision

MEASURE

Cohorts expanded only after approved thresholds

ACCEPTANCE EVIDENCE

Rollout plan with go/no-go gates

OBSERVED	THRESHOLD	DECISION	OWNER
machine value	approved rule	PASS / REPAIR	named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

12 · Feedback and graceful failure: execution path

1. Detect uncertainty
2. Explain limit
3. Offer recovery
4. Learn from feedback

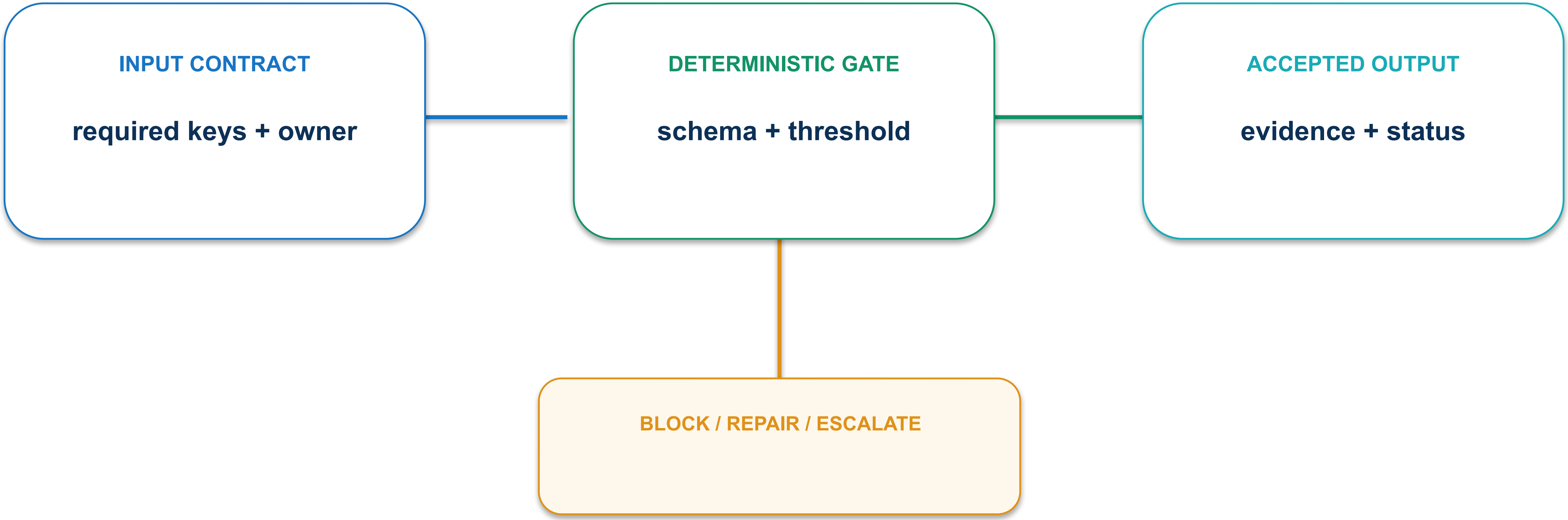
STATE PASSED BETWEEN STAGES

uncertainty	message
recovery	feedback

FAILURE MODE

The product hides uncertainty or traps users in repeated bad answers.

12 · Feedback and graceful failure: contract and control



Design honest limits, user control and a path to human help.

12 · Feedback and graceful failure: evidence and decision

MEASURE

Successful recovery rate after agent failure

ACCEPTANCE EVIDENCE

Failure-state prototype and feedback taxonomy

OBSERVED

machine value

THRESHOLD

approved rule

DECISION

PASS / REPAIR

OWNER

named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

MECHANISM

13 · Benefit-trade-off recommendation: execution path

1. Quantify benefit
2. Price risk
3. Test scenarios
4. Recommend posture

STATE PASSED BETWEEN STAGES

benefit	**risk**
scenario	**recommendation**

FAILURE MODE

The business case counts automation savings but excludes oversight, incidents and change cost.

13 · Benefit-trade-off recommendation: contract and control

INPUT CONTRACT

required keys + owner

DETERMINISTIC GATE

schema + threshold

ACCEPTED OUTPUT

evidence + status

BLOCK / REPAIR / ESCALATE

Use risk-adjusted scenarios and issue a pilot, expand, hold or stop decision.

13 · Benefit-trade-off recommendation: evidence and decision

MEASURE

Net value remains positive under downside scenarios

ACCEPTANCE EVIDENCE

Executive decision memo with conditions and owner

OBSERVED	THRESHOLD	DECISION	OWNER
machine value	approved rule	PASS / REPAIR	named human

Completion principle: evidence travels with the product hypothesis, changed state and decision—not in an unsupported claim.

Deployment boundary: product, platform and accountable owner

1

PRODUCT

Set value and experience target

2

PLATFORM

Operate agent and telemetry

3

OWNER

Approve rollout and exceptions

Adoption state: PILOT, EXPAND, HOLD and STOP

1. PILOT
2. EXPAND
3. HOLD
4. STOP

The adoption posture records evidence, conditions, owner and review date.

Production evidence chain: request → trace → metric → decision

request_id
user intent

trace_id
agent and tool actions

metric_set
quality and guardrails

decision_id
rollout posture

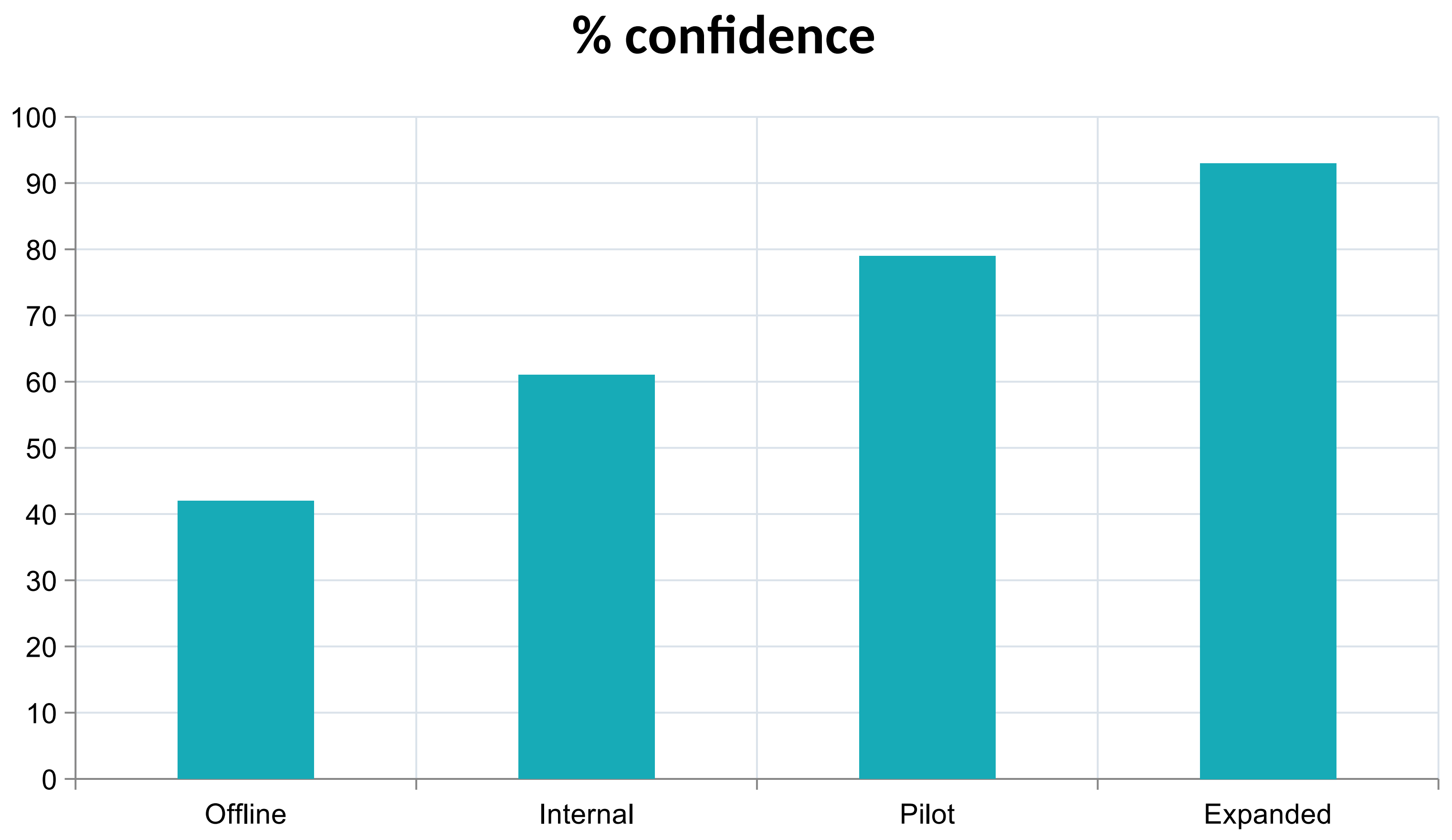
A production recommendation is defensible when every metric resolves to a trace and owner.

Deployment taxonomy: quality, reliability, cost and risk

QUALITY	repair causal behavior
RELIABILITY	repeat and segment trials
COST	optimise within guardrails
RISK	hold for accountable decision

EDITABLE NATIVE CHART

Staged rollout increases decision confidence



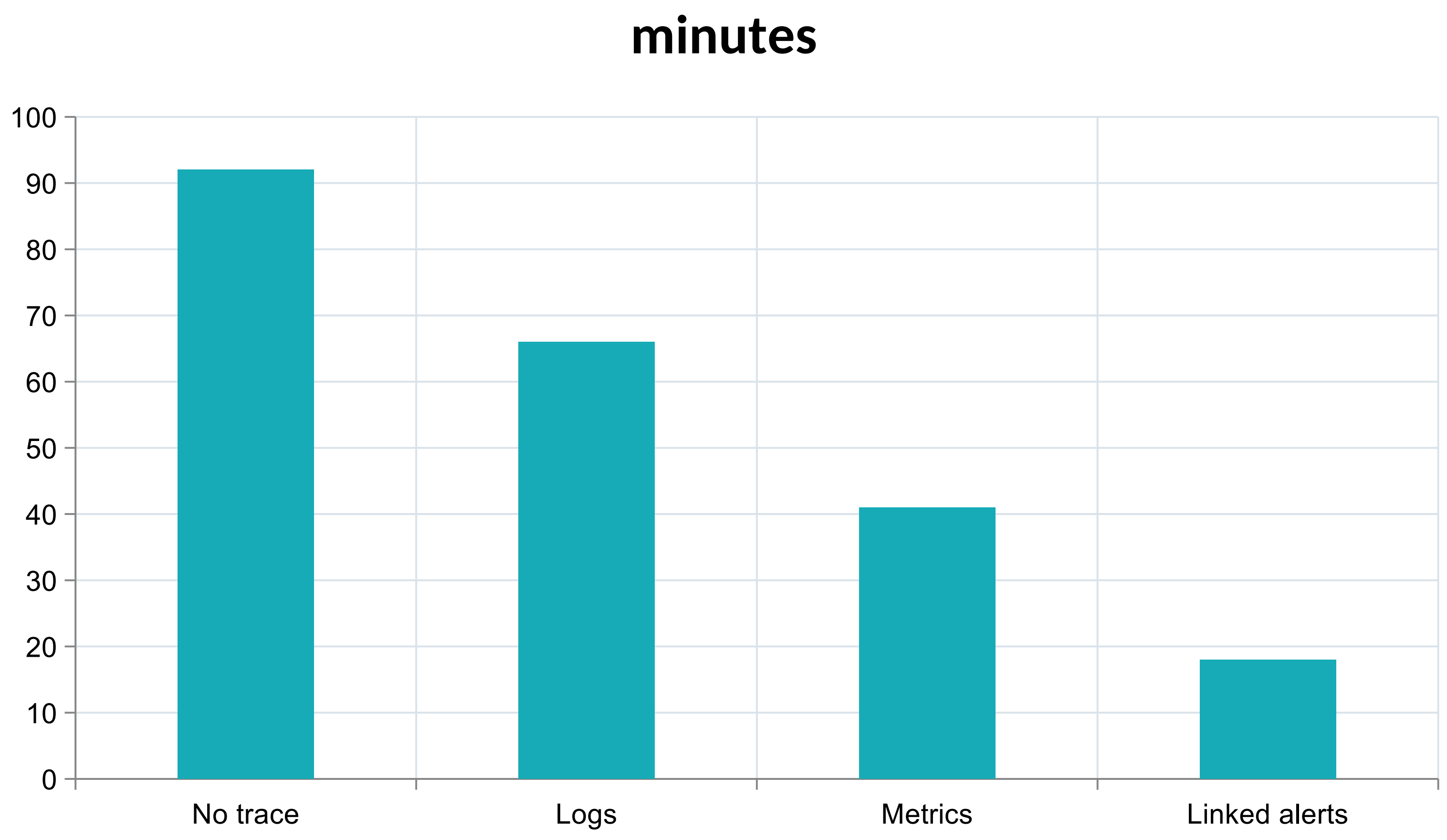
INSIGHT

Staged rollout converts production evidence into a safer expansion decision.

Illustrative synthetic values for mechanism explanation.

EDITABLE NATIVE CHART

Observability shortens recovery time

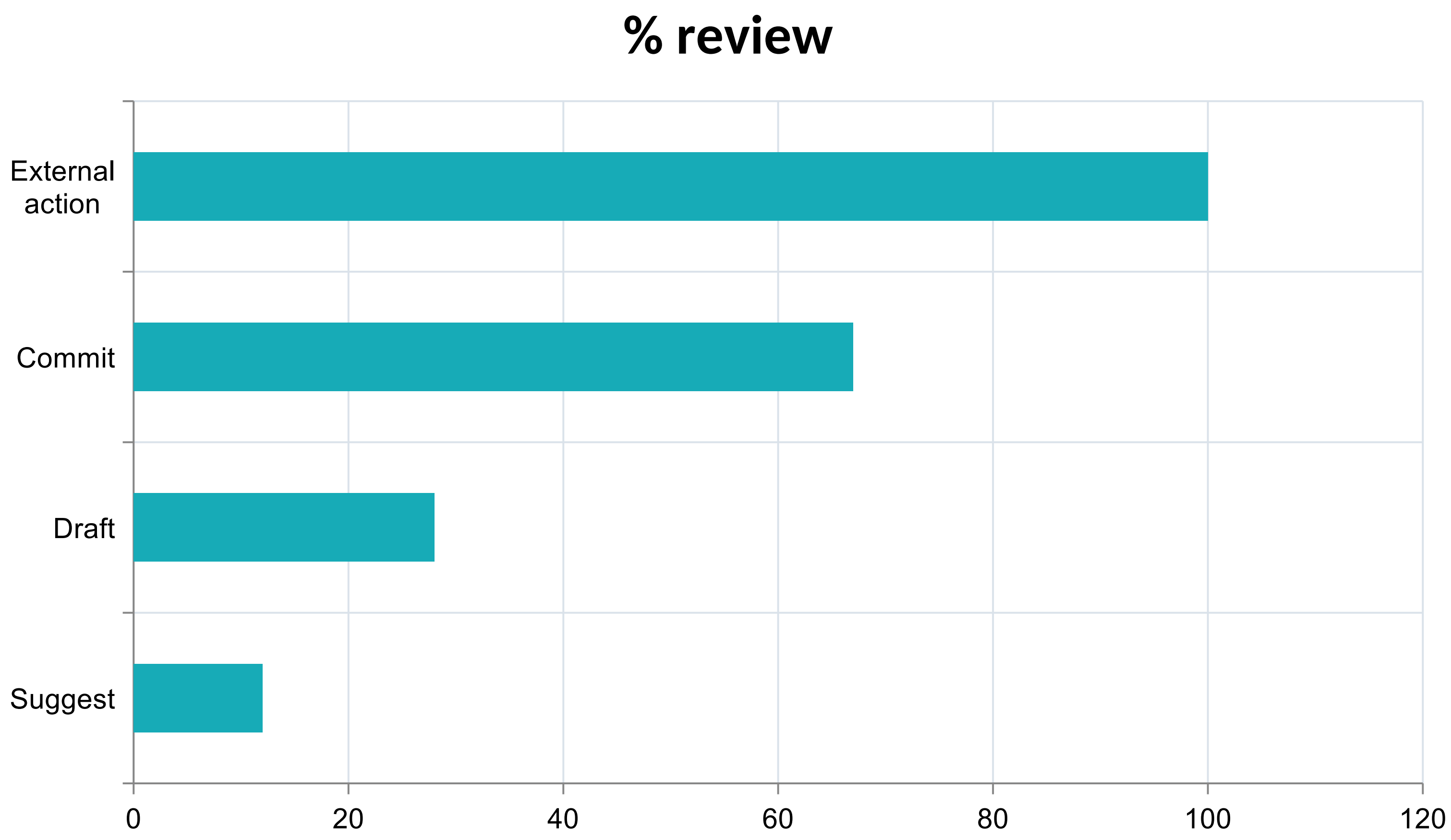


INSIGHT

Linked traces, metrics and alerts shorten the path from symptom to accountable action.

Illustrative synthetic values for mechanism explanation.

Human oversight rises with consequence



INSIGHT

Human oversight should increase with the consequence and reversibility of the action.

Illustrative synthetic values for mechanism explanation.

ACTIVITY BRIEF

Activity 09 · Select a Deployment Posture

SCENARIO

Compare channel and architecture options for an agentic product using task, integration and risk evidence.

EXPECTED OUTPUT

Deployment matrix, reference architecture and fallback path

Detailed procedures, prompts, troubleshooting and acceptance checks are in the Learner Guide and activity PDF.

Self-contained folder: activities/activity-09-deployment-decision/

Activity 09 · agent workflow contract

1. Map context
2. Compare channels
3. Select architecture
4. Add fallback
5. Cost posture
6. Approve

Consequential actions require a current diff, passing checks and a named decision.

HUMAN GATE

Approval resumes only when the reviewed diff and current state still match.

Activity 09 · evidence and failure gates

PASS EVIDENCE

The selected option meets task, integration, safety and recovery thresholds; caveats are explicit.

FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

README.pdf

detailed procedure

mock-data/

synthetic source state

templates/

learner working files

evidence/

acceptance record

ACTIVITY BRIEF

Activity 10 · Conduct a Risk and Control Review

SCENARIO

Threat-model an agent workflow and assign layered controls, evidence and owners.

EXPECTED OUTPUT

Risk register, permission map and control test plan

Detailed procedures, prompts, troubleshooting and acceptance checks are in the Learner Guide and activity PDF.

Self-contained folder: activities/activity-10-risk-control-review/

Activity 10 · agent workflow contract

1. Map assets
2. Identify threats
3. Score exposure
4. Select controls
5. Assign owner
6. Test control

Consequential actions require a current diff, passing checks and a named decision.

HUMAN GATE

Approval resumes only when the reviewed diff and current state still match.

Activity 10 · evidence and failure gates

PASS EVIDENCE

High risks have preventive and detective controls, named owners and testable evidence.

FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

README.pdf

detailed procedure

mock-data/

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acceptance record

Activity 11 · Design KPI and Feedback Dashboard

SCENARIO

Create a metric tree and monitoring view that balances business value, quality, experience, cost and risk.

EXPECTED OUTPUT

KPI tree, alert table and feedback taxonomy

Detailed procedures, prompts, troubleshooting and acceptance checks are in the Learner Guide and activity PDF.

Self-contained folder: activities/activity-11-kpi-feedback-dashboard/

WORKFLOW MAP

Activity 11 · agent workflow contract

1 Set outcome

2 Add guardrails

3 Define events

4 Set thresholds

5 Assign action

6 Review drift

Each node consumes named keys, emits a status, and leaves retrievable execution evidence.

Activity 11 · evidence and failure gates

PASS EVIDENCE

Metrics have formulas, data sources, thresholds, segments and action owners; no vanity-only measure drives release.

FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

README.pdf

detailed procedure

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acceptance record

ACTIVITY BRIEF

Activity 12 · Deliver a Governed Product Recommendation

SCENARIO

Synthesize value, evaluation, readiness and risk evidence into an executive pilot, expand, hold or stop decision.

EXPECTED OUTPUT

One-page decision memo and 90-day rollout scorecard

Detailed procedures, prompts, troubleshooting and acceptance checks are in the Learner Guide and activity PDF.

Self-contained folder: activities/activity-12-executive-recommendation/

Activity 12 · agent workflow contract

1. Summarise value
2. Compare evidence
3. Price risk
4. Test downside
5. Set conditions
6. Recommend

Consequential actions require a current diff, passing checks and a named decision.

HUMAN GATE

Approval resumes only when the reviewed diff and current state still match.

ACCEPTANCE MODEL

Activity 12 · evidence and failure gates

PASS EVIDENCE

The recommendation is traceable to evidence, survives a downside scenario, and has measurable exit criteria.

FAIL CLOSED WHEN

a protected boundary is crossed; evidence is missing; approval is absent; or the current diff no longer matches review.

README.pdf

detailed procedure

mock-data/

synthetic source state

templates/

learner working files

evidence/

acceptance record

CHECKPOINT

Topic 3 technical checkpoint

1. Which state field prevents a silent downstream assumption?

2. What evidence proves the mechanism completed successfully?

3. Which failure must stop rather than retry?

4. Who owns the consequential decision?

CHECKPOINT

Topic 3 transfer challenge

1. Change one input constraint and predict the affected nodes.
2. Name the identifier that preserves provenance across the change.
3. Define one machine threshold and one human decision.
4. Describe the minimum evidence required to resume the run.

Course achievement map

1. **LO1 Evaluate agentic AI builders for product research and development, comparing strengths, limitations, resources and skills.**
2. **LO2 Apply optimisation techniques to improve the efficiency, quality and control of agentic product-development workflows.**
3. **LO3 Evaluate the benefits and trade-offs of deploying agentic AI-powered product solutions and recommend a governed path forward.**

Access the learning platform

LMS

Use the LMS for attendance evidence, learner resources and assessment instructions.

1. **Open https://lms-tms.tertiaryinfotech.com/**
2. Use your assigned learner account
3. Confirm the course title and code before submitting evidence
4. Ask the trainer if access is blocked

Agentic product controls that must remain intact

GATE

Automation is complete only when governance travels with the product decision.

1. **Every agent role has a bounded goal and authority**
2. Every handoff preserves source and state provenance
3. Every consequential action has a named human owner
4. Every release claim resolves to evaluation evidence

Assessment reminder

A

Demonstrate understanding and an evidence-backed end-to-end workflow.

1. **Written Assessment (SAQ): 6 questions / 60 minutes**
2. Practical Performance: 3 tasks / 60 minutes
3. Use only learner-safe activity data and product evidence
4. Submit through the instructed LMS route

Assessment flow

1→4

Complete both components and retain the requested evidence.

1. **Briefing and identity check**
2. Written Assessment (K1–K6)
3. Practical Performance (A1–A3)
4. Assessor review and outcome

Digital attendance is mandatory

✓

Attendance evidence is part of the course administration record.

1. **Check in at the instructed time**
2. Use the correct learner identity
3. Complete both attendance events each training day
4. Notify the trainer immediately if the form fails

References and further study

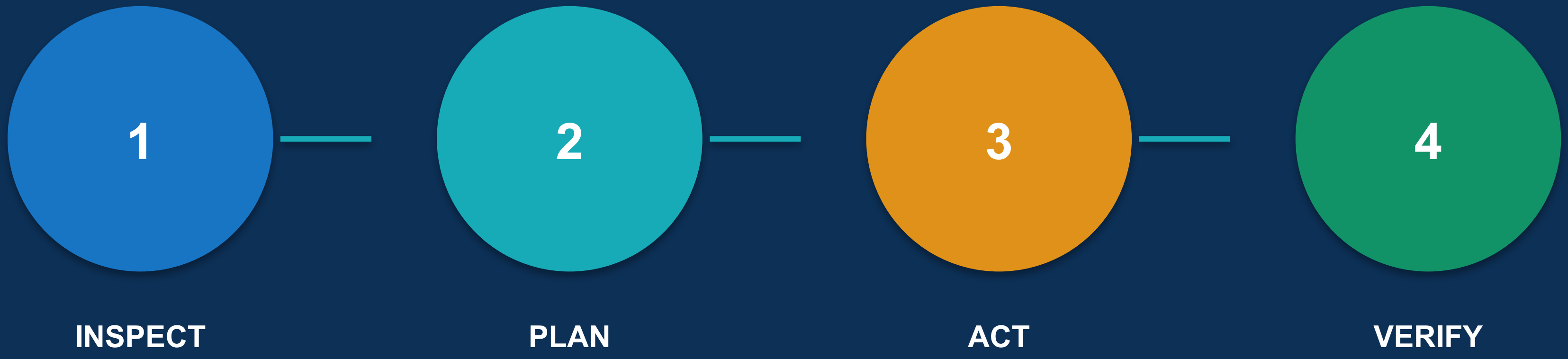
REF

The course synthesises the original deck with both supplied eBooks and current primary guidance.

1. **AI Product Management — Aman Khan**
2. Building AI-Powered Products — Marily Nika
3. Anthropic — Building effective agents
4. Google PAIR — People + AI Guidebook
5. NIST — AI Risk Management Framework
6. Tertiary Courses — current course specification

Build agentic products that can prove what happened.

Evidence before completion • Human approval before consequence



Dr Alfred Ang | Tertiary Infotech Academy Pte. Ltd.

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